Positional Information for Budget-Adaptive Markov Prediction

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Abstract

We study next-symbol prediction from one binary order-two Markov trajectory after an unknown number of symbol edits, with the terminal pair protected. Loss is expected total variation at that actual pair. At a supplied all-state two-step minorization certificate $1/M$, the known-budget risk scale is $g_M(k) = \max\{M^{-1/2}, k/M\}$. For every $M \geq 64$ and $2 \leq r \leq M+2$, we prove that the optimal competitive factor of one budget-blind map observing exact triple counts and the first and last r symbols has order $1 + \log((M+2)/r)$, with explicit universal constants. The upper uses only counts and the suffix; the lower grants both windows and compares the same map across budgets. A whole-word scan can instead supply one exact positional gate bit beyond counts and the protected pair. We then give a perturbation bound for approximate full and half-prefix counts, a fully accounted finite masked-attention/Boolean-ReLU realization, and a matching clean precision bound for its twelve prescribed rounded coordinates. These latter results calibrate a specific observation interface: they do not establish neural superiority over same-information counters, a learning result, or a precision law for an entire network. Complete proofs and the distinctions among the observation models are included.

1 The observation question

A transition histogram records which local patterns occurred but not when they occurred. For a clean homogeneous Markov trajectory, this is consistent with the usual likelihood factorization into transition counts and the initial state. With position-sensitive corruption, however, the observation channel changes. A predictor may need to distinguish an empirical row supported by older visits from one populated almost entirely near the prediction time. The question here is how much positional information is needed when the number of edits is not supplied.

We study one concrete experiment: a binary order-two source, M observed departures, and an unmodified terminal pair. The source obeys an all-state two-step minorization with supplied coefficient $1/M$. An offline kernel may change at most k earlier symbols on every realization. The predictor sees exact triple counts and prescribed boundary windows, but not k. Its risk is expected unsquared total variation to the next-symbol distribution at the actual terminal pair. In this experiment, giving k to the predictor makes counts and the protected pair sufficient for the optimal *rate*; no claim of statistical sufficiency is needed.

Our main result characterizes the competitive factor simultaneously for every finite horizon and boundary width: its order is $1 + \log((M+2)/r)$ with universal constants. One map at each fixed horizon must serve every budget. The positive lower limiting fraction criterion and logarithmic fixed-fraction price follow at the order level; a separate fixed-fraction proof supplies a stronger explicit lower constant. A separate corollary shows why raw access and computed messages must not be identified: a whole-word scan can produce a single exact gate bit that suffices beyond counts. The scan, exact counters, and protected query are part of that corollary's information model.

The finite implementation is subordinate to this information question. A discontinuous accept/reject gate need not be recovered exactly: uniform approximation of its full and half-prefix statistics controls expected prediction error. An explicit masked-attention program realizes these statistics, while a classical one-pass counter realizes the same information more simply. We keep these computational comparisons separate from the restricted-observation lower bound.

The proof mechanisms are complementary. For the upper bound, a gate retains evidence of visits before a suffix, removing the shortest-return terms from a harmonic convolution. For the lower bound, clean alternatives with small prediction risk are transformed into legally edited null words whose restricted observations match those alternatives. The clean guarantee is used at the same horizon and supplied certificate as the attacked guarantee. Complete arguments appear in the appendices. Sections 2–5 give the statistical comparison, Section 6 calibrates one finite implementation, and Section 7 states the relationship to prior work.

2 Experiment and benchmark

All logarithms are natural. Let $M \geq 64$ be an integer and

$$X = (X_0, \ldots, X_{M+1}), \qquad Y_i = (X_i, X_{i+1}), \quad 0 \leq i \leq M.$$

Thus there are $M + 2$ raw symbols, M departures, and the terminal pair Y_M. A homogeneous order-two transition table $\theta_{ab}(c)$ induces the four-state pair kernel $P((a, b), (b, c)) = \theta_{ab}(c)$. There is a qualifying stationary law Π such that

$$\Pi P = \Pi, \qquad P^2(u, v) \geq \beta \Pi(v) \quad \text{for all } u, v \in \{00, 01, 10, 11\}. \tag{2.1}$$

The condition includes rows outside stationary support. Throughout the main results the supplied certificate is *exactly* $\beta = 1/M$; it need not be the source's largest possible coefficient. Neither θ nor Π is revealed. Stationary initialization means $Y_0 \sim \Pi$; arbitrary initialization means any hidden law μ, including laws supported outside the stationary support.

Strict edits and actual-terminal loss. For $0 \leq k \leq M$, a legal randomized kernel $K_{M,k}$ maps full binary words to full binary words. On every input x, every output z in its support satisfies

$$z_M = x_M, \qquad z_{M+1} = x_{M+1}, \qquad \sum_{i=0}^{M-1} \mathbf{1}\{z_i \neq x_i\} \leq k. \tag{2.2}$$

The kernel is one fixed map on words, not supplied the true source table, stationary or initial law, or predictor's random seed. It may inspect the whole input and use public parameters and k; lower-bound kernels may hardcode a predeclared auxiliary family. No causal or computational restriction is imposed. Its randomness is independent of the predictor's. This formulation specifies the common kernels used in the proofs; source independence alone is not asserted to distinguish all standard contamination models.

For $Z \sim K_{M,k}(X, \cdot)$ and an estimator A based on observation $\mathcal{T}(Z)$, define

$$R(A; \theta, \mu, K_{M,k}) = \mathbb{E}\ \mathrm{TV}\big(A(\mathcal{T}(Z)), \theta_{Y_M}\big). \tag{2.3}$$

The expectation includes the path and both independent randomizations. The target is the source's next-symbol law at the *actual protected pair*, not an independent query. For binary distributions, TV is absolute error in either coordinate. Risk is unconditional.

Observed information. For $a, b, c \in \{0, 1\}$ let

$$H_{abc}(z) = \sum_{i=0}^{M-1} \mathbf{1}\{(z_i, z_{i+1}, z_{i+2}) = (a, b, c)\}.$$

For prescribed $2 \le r \le M + 2$, the observation is

$$\mathcal{T}_r(z) = \big((H_{abc}(z))_{a,b,c},\ z_0, \dots, z_{r-1},\ z_{M+2-r}, \dots, z_{M+1}\big). \tag{2.4}$$

The windows may overlap. Counts and raw symbols are exact; neither edit locations nor the clean interior are supplied. In row $u = (a, b)$ use the designated successor $d(u) = 1 - a$, and define

$$J'_u = H_{ab0}(Z) + H_{ab1}(Z), \qquad C'_u = H_{ab,d(u)}(Z), \qquad q_u = \theta_{ab}(d(u)).$$

Primes indicate corrupted counts. Unprimed counts are clean analysis variables. At the selected row $u = Y_M$, subscripts are omitted. A ratio C/J is evaluated only when $J > 0$.

One family for all budgets. Use the explicitly normalized rate profile

$$g_M(k) = \max\{M^{-1/2}, k/M\}, \qquad 0 \le k \le M. \tag{2.5}$$

For a prescribed sequence $r_\bullet = (r_M)_{M \ge 64}$, let $C_{\mathrm{stat}}(r_\bullet)$ be the infimum of constants $C \ge 0$ for which one estimator family (A_M), based on $\mathcal{T}_{r_M}$, satisfies

$$R(A_M; \theta, \Pi, K_{M,k}) \le C g_M(k) \tag{2.6}$$

for every $M \ge 64$, every $0 \le k \le M$, every qualifying source and every legal kernel. Define C_{all} by additionally quantifying over every μ. The family may depend on the prescribed sequence, M, r_M, and supplied $1/M$, but not on k or hidden source quantities. The infimum need not be attained; an empty admissible set has value $+\infty$.

Finite-horizon competitive factors. Separately from the sequence factors C, fix (M, r) and a single k-blind map A based on $\mathcal{T}_r$. Define

$$\Gamma_{\mathrm{stat}}(A; M, r) = \sup_{\substack{0 \le k \le M,\ \theta, \Pi, K_{M,k} \\ \text{qualifying stationary experiments}}} \frac{R(A; \theta, \Pi, K_{M,k})}{g_M(k)}, \qquad \Gamma_{\mathrm{stat}}(M, r) = \inf_A \Gamma_{\mathrm{stat}}(A; M, r). \tag{2.7}$$

Define $\Gamma_{\mathrm{all}}(A; M, r)$ and $\Gamma_{\mathrm{all}}(M, r)$ by also allowing every initial law μ. The map may depend on M, r and the supplied $1/M$, but the very same map serves all k at that horizon. These are finite-horizon infima, not the sequence quantities C above; no infimum is assumed attained. Every map has factor at most $\sqrt{M}$ because TV is at most one.

The comparison is to a genuine same-information known-budget scale, rather than to an arbitrary denominator. Write $R^{\mathcal{T}}_{\mathrm{stat}}(M, k)$ and $R^{\mathcal{T}}_{\mathrm{all}}(M, k)$ for the respective minimax risks when k is also supplied.

Proposition 2.1 (Known-budget scale)**.** *For the full corrupted history, for $\mathcal{T}_r$, and for counts plus the protected pair,*

$$\frac{g_M(k)}{96} \le R^{\mathcal{T}}_{\mathrm{stat}}(M, k) \le R^{\mathcal{T}}_{\mathrm{all}}(M, k) \le 191488 g_M(k). \tag{2.8}$$

The upper is count-only and allows arbitrary starts; the lower holds even for full histories. Thus a factor C relative to g gives a factor at most $96C$ relative to the corresponding minimax oracle. Conversely, a factor D relative to that oracle gives a factor at most $191488D$ relative to g. Proofs are in Lemma B.5 and Proposition A.10.

For comparison with a more general certificate-dependent profile, define $F_2(M, 0, \beta) = 0$ and, for $k \geq 1$,

$$F_2(M, k, \beta) = \frac{k}{M}\left[1 + \log\left(1 + \frac{1}{k \max\{\sqrt{\beta}, 1/M\}}\right)\right], \qquad H_2 = \min\{1, \max\{M^{-1/2}, F_2\}\}.$$

At the certificate used here, $\log(1+x) \leq x$ gives $g_M(k) \leq H_2(M, k, 1/M) \leq 2g_M(k)$. Hence the two normalizations change optimal factors by at most two; all constants below are stated directly for g.

3 Finite-horizon characterization and its consequences

Theorem 3.1 (Simultaneous finite-horizon window price)**.** *For every integer $M \geq 64$ and $2 \leq r \leq M + 2$, put $\ell = 1 + \log((M+2)/r)$. Then*

$$2^{-30}\ell \leq \Gamma_{\mathrm{stat}}(M, r) \leq \Gamma_{\mathrm{all}}(M, r) \leq 2^{20}\ell. \tag{3.1}$$

The upper uses only exact counts and the suffix. The lower grants both boundary windows. All experiments use the supplied certificate $1/M$.

The proof in Section C checks the complete finite domain of the matched-observation bound. It tests the clean risk of the same map before selecting an attacked budget; it is not a logarithmic risk lower at each budget. Linear widths have constant factor; for fixed $0 < \gamma < 1$, $r = \max\{2, \lceil M^\gamma \rceil\}$ has order $\log M$, as does $r = 2$. The constants are conservative, not a claim of sharp practical thresholds.

The next sequence statements are consequences at the order level. Their original separate proofs are retained in Proposition B.4, including the stronger fixed-fraction lower constant.

Theorem 3.2 (Exact raw-window criterion)**.** *For every prescribed sequence of integers $2 \leq r_M \leq M + 2$,*

$$C_{\mathrm{stat}}(r_\bullet) < \infty \iff C_{\mathrm{all}}(r_\bullet) < \infty \iff \liminf_{M\to\infty} \frac{r_M}{M} > 0.$$

No monotonicity of r_M is required. The factor may depend on the prescribed sequence.

Necessity already uses stationary witnesses. Sufficiency includes unsupported starts. A finite initial set of horizons is harmless because $g_M(k) \geq M^{-1/2}$. The conclusion is about one budget-blind family, not a separate minimax estimator chosen for each k.

Theorem 3.3 (Price of a fixed fraction)**.** *Fix $0 < \alpha \leq 1/4096$ and $r_M = \max\{2, \lceil \alpha M \rceil\}$. Abbreviating the corresponding factors by $C_{\mathrm{stat}}(\alpha)$ and $C_{\mathrm{all}}(\alpha)$,*

$$2^{-26} \log \frac{1}{32\alpha} \leq C_{\mathrm{stat}}(\alpha) \leq C_{\mathrm{all}}(\alpha) \leq 2^{19}\left(1 + \log \frac{2}{\alpha}\right). \tag{3.2}$$

The upper requires only the histogram and suffix; the lower permits both boundary windows.

Thus the optimal uniform factor has order $\log(1/\alpha)$ as $\alpha \downarrow 0$. The upper is uniform in every M, k for each fixed α. The lower first sends $M \to \infty$ with α fixed and with a proposed finite factor fixed. This fixed-fraction proof takes limits in the stated order. The finite characterization above requires no limit in either parameter; neither result is a logarithmic risk lower bound at every budget. The constants are explicit but not optimized. Proposition B.4 proves Theorems 3.2 and 3.3 from a finite upper bound and a finite matched-observation lower bound.

3.1 An explicit estimator

Set

$$m = \min\{r-2, \lfloor M/2 \rfloor\}, \qquad N = M - m, \qquad L = 1 + \log \frac{M}{m+1}. \tag{3.3}$$

The last $m+2$ symbols reveal exactly the m departures indexed $M-m, \dots, M-1$ and the terminal pair. Let T'_u count visits to u at those departures and put $O'_u = J'_u - T'_u$. Then O'_u is the count on indices $0, \dots, N-1$; it is a direct old-prefix count even though it is computed by subtraction. At the selected row, predict designated probability

$$\widehat{q} = \begin{cases} C'/J', & J' > 0 \text{ and } 512O' \geq J', \\ 0, & \text{otherwise.} \end{cases} \tag{3.4}$$

The complementary symbol receives probability $1 - \widehat{q}$. This map does not use k or β. When $r = 2$, $m = 0$ and $O' = J'$; the terminal pair is not a departure. Using $r-1$ in place of $r-2$ would overcount the available information.

The finite upper bound at the main certificate, proved in Proposition A.9, is

$$R \leq \min\left\{1, \frac{131072}{\sqrt{M}} + (8448L + 147744)\frac{k}{M}\right\} \leq 287264 L g_M(k). \tag{3.5}$$

If r_M/M is eventually bounded below, then L is eventually bounded. For fixed α as in Theorem 3.3, $m + 1 \geq \alpha M/2$, including the horizons at which $r_M = 2$. This proves the two stated upper conclusions without a normalization loss.

4 Why positions enter the proof

Older visits and a truncated harmonic sum. The gate is not an estimate of the budget. It requires a positive fraction of the selected row's visits to precede the observed suffix, not merely one old occurrence. This proportional balance makes gate-switch errors chargeable to the edit budget; one forged old occurrence alone must not unlock a large corrupted suffix count. For a clean d-state homogeneous chain and any start, the relevant occupation inequality is

$$\mathbb{E}\frac{\mathbf{1}\{O \geq cJ\}}{J+1} \leq (3d^2 + 2d^3)\frac{1 + \log(M/(m+1))}{M}, \qquad c > 0. \tag{4.1}$$

Here J is the number of departures from the actual terminal state and O counts those before time $M - m$. Lemma A.7 proves this by an exact first-hit convolution and a reciprocal-return identity. In this analytic bound the event can be enlarged to the existence of an old visit. That enlargement removes short backward-time terms; the surviving harmonic range produces the logarithm. The implemented gate still requires proportional balance. Generic finite-state first-hit pointmass estimates have precedents in Norris et al. [18]; the required gated convolution is proved explicitly.

A strict k-edit outcome changes each relevant count by at most $6k$. A deterministic true-gate/false-gate split therefore reduces the error to a clean empirical comparator, the reciprocal count in (4.1), a weighted imbalance term, and two small-occupation events. The all-state certificate controls the imbalance even from unsupported starts. No conditional independence of departures given the random terminal row or its occupation is assumed. The supplement's row-stack argument instead starts from independent row-wise randomness and then bounds the selected empirical deviation.

Mixture domination and legal observation matching. The lower bound uses stationary alternatives whose designated exit probabilities are

$$(q_{00}, q_{01}, q_{10}, q_{11}) = (a, 1/2, 1/2, t), \qquad \Pi_a = \frac{(t/a, 2t, 2t, 1)}{1 + t/a + 4t}.$$

An integral over a creates the logarithmic term. Conditioning on a fixed departure count, one counts marked excursions and holding slots; the resulting occupation-mixture bound controls each proposed output mass. An interior loop relocation preserves the alternative's triple histogram and both of its observed windows, while making it close to a simple two-run null word. These are the alternative's windows, not windows of the unedited null input: changing an observed input-window symbol is legal but must be charged to the budget.

The attack normalizes only nonzero proposal masses, samples once, performs the relocation, and checks the actual Hamming cost and terminal pair. If any check fails, it returns its input unchanged. It is therefore a common source-independent kernel satisfying the budget on every outcome, not only on a high-probability event. In the risk comparison, nonnegative loss is discarded before proposal-mass domination; a signed triangle inequality is used only afterwards. Lemmas B.1 and B.2 and Proposition B.3 supply these steps.

For sublinear widths along any subsequence, a budget between the window and horizon scales makes the clean risk negligible relative to the alternative's exit scale while the mixture logarithm diverges. For fixed α, a budget proportional to the window yields $\log(1/\alpha)$. The alternatives may depend on this tested budget, but their clean risk is bounded using the same estimator at the same M and supplied $1/M$. This constrained-risk logic is related to general risk-transfer arguments such as Duchi and Ruan [6]; the dependent-word construction and legal observation matching are additional obligations.

5 One computed positional bit

Now change the access model. The decoder receives exact triple counts and the protected pair, optionally also the initial pair. An encoder may scan the entire already corrupted word and send one symbol in a fixed finite alphabet $\mathcal{A}$. It knows M but not k, β, or the source. The message is exact and fault-free after construction. There are no timing, framing, variable-length, or other side channels, and no shared source-dependent information.

Corollary 5.1 (Minimum supplemental alphabet)**.** *In this scan-and-message model, the minimum $|\mathcal{A}|$ for a finite uniform factor relative to $g_M(k)$ at supplied $1/M$ is two. This holds for stationary or arbitrary initialization, with or without the initial pair supplied.*

For the upper, put $n = \lfloor M/2 \rfloor$, select $u = Y_M$, and send

$$B = \mathbf{1}\{512J_1' \geq J'\}, \qquad J_1' = \sum_{i=0}^{n-1} \mathbf{1}\{(Z_i, Z_{i+1}) = u\}. \tag{5.1}$$

The decoder outputs C'/J' if $J' > 0$ and $B = 1$, and zero otherwise. Equality is accepted. If $J' = 0$, the bit equals one, but the explicit positive-denominator test enforces the fallback. Proposition A.9 gives all-start risk at most $\min\{1, 2^{17}M^{-1/2} + 2^{23}k/M\}$. A one-symbol alphabet supplies no information. Theorem 3.2 with $r_M = 2$ rules it out even when both endpoint pairs are present. This proves minimality without another lower-bound construction.

A finite word illustration, not a risk experiment. For $M = 65$, the two words in Table 1 have the same initial pair 01, terminal pair 10, and triple counts, in lexicographic order,

$$(H_{000}, H_{001}, H_{010}, H_{011}, H_{100}, H_{101}, H_{110}, H_{111}) = (0, 1, 1, 1, 1, 0, 1, 60).$$

At row 10, $J' = C' = 1$. The single departure from that row lies in different halves, so the computed gate differs although $\mathcal{T}_2$ is identical. The example verifies a combinatorial collision only: it gives no source probabilities, statistical risk comparison, or lower-bound proof.

Table 1: An exact 67-symbol collision. The notation 1^{62} means 62 consecutive ones; the first half contains departures 0 through 31.

Word	J'_1	B	Designated prediction
$0100\,1^{62}\,0$	1	1	1
$0\,1^{62}\,0010$	0	0	0

This is a worst-case supplemental alphabet result, not an entropy or mutual-information statement. The baseline exact counts are uncharged inputs. One computed bit is not one observed raw coordinate and does not remove the whole-word scan. A counter-based implementation is a natural baseline: maintain the eight triple counts and four first-half row counts, retain the endpoint, and select its row after the scan. This uses $O(\log M)$ counter workspace, $O(M)$ counter updates, and at most $O(M \log M)$ naive bit operations. No neural mechanism is needed for this statistical rate guarantee.

6 Approximate statistics and a finite realization

The positional gain is a property of information, not of an attention architecture. We now fix one finite representation of that information and distinguish three questions: stability of the statistical decoder, implementation of the representation, and the precision required by an arbitrary decoder of that fixed representation. Throughout this section h is bank precision and b is output precision; r retains its meaning as raw-window width.

6.1 A direct full and half-prefix bank

For every row u, retain the full counts J'_u, C'_u already defined, and let

$$D'_u = \sum_{i=0}^{\lfloor M/2 \rfloor - 1} \mathbf{1}\{(Z_i, Z_{i+1}) = u\}.$$

Thus D'_u/M is a subfrequency normalized by M, not by the half length. The twelve direct coordinates are

$$(J'_u/M, C'_u/M, D'_u/M)_{u \in \{00,01,10,11\}}. \tag{6.1}$$

The decoder additionally receives the exact protected terminal pair. No other historical information is implicit in this bank.

Suppose a computation approximates each unnormalized coordinate within $e \geq 0$ on every allowed outcome. The errors may be correlated, real-valued, and inconsistent with any word. Clip

the total to $[0, M]$, then the successor and half count to the resulting interval $[0, \bar{J}_u]$. At the selected row, use

$$\widehat{q} = \begin{cases} \bar{C}/\bar{J}, & \bar{J} > 0 \text{ and } 512\bar{D} \geq \bar{J}, \\ 0, & \text{otherwise.} \end{cases} \tag{6.2}$$

Sequential clipping preserves coordinate error, including when the total rounds to zero. Theorem D.1 proves that any further probability approximation within ξ yields, at the critical certificate,

$$R \leq \min\left\{1, \frac{131072}{\sqrt{M}} + 6433056\frac{k}{M} + 1072176\frac{e}{M} + \xi\right\}. \tag{6.3}$$

The single decoder is independent of k, e, and ξ. The appendix also states the general positive-certificate bound.

This stability statement does not require a margin separating the gate from equality. On clean rows of large occupation, an approximate false acceptance or rejection is incompatible with the gap between the analytic balance thresholds $1/1024$ and $1/256$. The remaining gate changes occur on small clean rows and are paid by the actual-terminal occupation tail. The proof handles real error radii with integer ceilings; it does not condition the row departures into independent samples.

6.2 A paid masked-attention implementation

Set $Q = 2^h$ with $h \geq 0$. Round each coordinate of (6.1) separately to the nearest point of $\{0, 1/Q, \ldots, 1\}$, with ties upward. Let its integer numerator be $B_i \in \{0, \ldots, Q\}$. The represented endpoint is exact. The allowed numeric fault replaces the twelve master numerators jointly by canonical integers B_i^* in the same range, satisfying

$$\max_i |B_i^*/Q - B_i/Q| \leq \delta.$$

Only those replaced values, the endpoint, and public source-independent format metadata are available downstream. Endpoints, masks, addresses, and the subsequent arithmetic remain exact. This is a post-rounding coordinate replacement model, not arbitrary analog noise or register bit flips.

Theorem F.1 constructs the bank for every binary word. Two zero-score singleton-mask heads copy the preceding one and two bits. At departure key $i + 2$, tokenwise Boolean/ReLU circuits form row, designated-successor, and first-half indicators. A third zero-score head at a synthetic final query averages exactly the M departure keys. Its finite evaluator accumulates integers without intermediate rounding, then rounds once. The same copy layer acquires the terminal pair. All structural masks, positions, local features, storage, and copying are charged.

A finite comparator implements equality acceptance in (6.2); restoring division produces an output with error at most 2^{-b}. The construction covers empty rows, $h = 0$, $b = 0$, and numerator $B_i = Q$, whose binary representation requires $h + 1$ bits. If the bank is presented as scalars rather than packed bits, explicit finite-grid ReLU extraction is included rather than treating scalar-to-bit conversion as free wiring. The resulting risk is

$$R \leq \min\left\{1, \frac{131072}{\sqrt{M}} + 6433056\frac{k}{M} + 1072176(2^{-h-1} + \delta) + 2^{-b}\right\}. \tag{6.4}$$

The predictor does not receive k or δ. This is a composition of a deterministic all-word construction with (6.3); it is not a training result.

What the small hybrid shape counts. Put $n = M+3$, $\ell_a = \lceil \log_2(n+1) \rceil$, and $p = \ell_a + h + b + 32$. The hybrid has two attention layers and three heads in total, value width at most twelve, and shared pointwise Boolean/ReLU depth $O(p^2)$ and width $O(p)$. Its attention primitive is evaluated by paid finite integer arithmetic. Table 2 summarizes two complete, alternative implementations; their costs cannot be mixed to select the smallest entry in each row. All-token pointwise execution, holds, zero coefficients in the dense version, input, endpoint acquisition, and clearing are included. Appendix F supplies offline construction and the full ledgers.

Table 2: Conservative finite implementation bounds. Each column is one complete representation. Peak storage uses the conservative scalar allocation and includes simultaneously live code, masks, input, and work arrays.

Resource	Sparse lists and circuits	Dense masks and circuits
Online bit operations	$O(np^4)$	$O(n^2p^2 + np^6)$
Program and code bits	$O(p^4)$	$O(p^5)$
Mask and position bits	$O(n\ell_a)$	$O(n^2 + n\ell_a)$
Total peak bits	$O(p^4 + n\ell_a + np^2)$	$O(p^5 + n^2 + n\ell_a + np^2)$

A pure Boolean/ReLU network can instead unroll the entire bit-operation schedule. With its paid time T and total live storage S, the resulting depth is $O(T + p)$, width $O(S + p)$, and sparse gates and holds $O((T + p)(S + p))$. These are not the smaller hybrid depth and width. The bank's $12(h + 1) + 2$ transmitted bits are likewise not the program's total memory.

The same-information counter comparison. A classical one-pass algorithm maintains four triples of exact J, C, D counters and a two-bit sliding window, updating the half count only before $\lfloor M/2 \rfloor$. It retains the final pair and can round once at the end to produce exactly the same bank. Its auxiliary storage is $O(\log M)$ bits and conservative counting work is $O(M \log M)$ bit operations, plus separately paid finite output division. Resident input costs $M + 2$ additional bits; a read-once input assumption must be stated if that storage is omitted. This baseline achieves the exact-statistic rate and is computationally simpler than the attention realization. The comparison to count-only prediction isolates the added positional information, not an advantage of neural computation.

6.3 Precision of the fixed bank

Let $R^*(M, h)$ be the clean minimax actual-terminal TV risk when the only observation is the twelve separately nearest-rounded coordinates in (6.1), the exact endpoint, and source-independent public metadata. The decoder may have arbitrary computation, real-valued probability outputs, and independent randomization. The source class is the same critical-certificate class with arbitrary starts, but no source edits or numeric faults. Theorem E.1 proves

$$\frac{1}{1600000} \max\{M^{-1/2}, 2^{-h}\} \le R^*(M, h) \le \min\{1, 667160 \max\{M^{-1/2}, 2^{-h}\}\}. \tag{6.5}$$

Both lower terms hold on the stationary IID Bernoulli subclass. The precision proof constructs two separated parameters whose twelve mean curves share a rounding cell, proves concentration of the actual random bank in that cell, and couples the retained endpoint without asserting independence from the bank. A full-history two-point bound covers the complementary sample-size regime.

Consequently $h = \frac{1}{2} \log_2 M + O(1)$ is the scale needed to retain a fixed multiple of the clean statistical rate for this particular bank. This is not a bound on all encoders with the same total

number of bits. It is not a lower bound on the arithmetic precision, addresses, accumulator width, parameters, scores, or total storage of a Transformer. The clean lower also does not assert matching δ or k/M terms for this new fault port. Equation (6.4) is the proved robust upper.

7 Relation to prior work and scope

Clean positional prediction. Han et al. [10, model (1), Theorem 24, and Section VI.A] study stationary prediction at the actual terminal state, including suffix-based estimators and higher-order KL analysis. Theorem 24 gives the binary order-two KL result; the squared-TV result in Section VI.A concerns first-order chains with fixed state count at least three. These are direct antecedents, not independent-query results. Their clean full-trajectory experiment differs from the strict-edit and restricted-observation comparison here. Losses must also be distinguished: a squared-TV upper bound implies an unsquared-TV upper bound by Jensen, but a squared-TV lower bound does not yield its square root as an unsquared-TV lower bound. The spectral-gap-restricted results in that work use an assumption different from the supplied all-state minorization (2.1).

Unknown contamination. Budget adaptation is not new in itself. Jain et al. [11] select among candidate estimates with monotone radii under a simultaneous-validity premise: every valid-index ball must contain the target. Separate expected-risk bounds do not establish that premise. Their selection lemma is pointwise, however, so the random terminal target is not by itself an obstruction: it could be applied realization by realization if all candidate predictions share the protected query and simultaneous accuracy is proved. Bhatt et al. [3] explicitly construct confidence-based adaptation for a mean under strict retrospective replacements of IID observations. Chen et al. [4] give robust testing and estimation under IID Huber mixtures, while Liu and Gao [15] exhibit an adaptation obstruction involving both smoothness and contamination. None of these comparisons licenses a general claim that unknown contamination alone prevents adaptation.

Dependence and count-based inference. Maity and Mitra [16, Theorem 2] analyze robust mean estimation for a stationary finite ergodic Markov chain as a tool for temporal-difference learning. Their procedure takes a corruption fraction and mixing time; Bernoulli corruption indicators replace rewards rather than edit the state trajectory. Its invariant-mean target and mixing-dependent confidence bound differ from the actual-terminal word experiment here. Robustness under dependence is therefore an established subject, not itself the proposed contribution.

Lecestre [12] estimate marginal or invariant distributions under dependence; their contamination corollary imposes independence conditions on the replacement channel. Leskelä and Dreveton [14] estimate transition matrices from independent parallel trajectories with arbitrarily corrupted whole trajectories. These are not the one-word terminal-target experiment. They also show why the present result should not be summarized as a universal failure of counts under corruption.

Clean transition-count factorization and Markov exchangeability remain valid. The report of Diaconis and Freedman [5] and the finite-stack discussion of Halberstam and Hutchcroft [9] provide relevant background on count-compatible paths. For a clean order-two word, likelihood is its initial-pair mass times transition probabilities raised to triple counts. After a position-sensitive kernel K, an observed word's probability instead sums these clean probabilities weighted by $K(z \mid x)$. Clean factorization alone does not establish sufficiency in this altered experiment. The specific observation map, not an unrestricted claim about all compact summaries, is essential.

Limits of the conclusions. The lower bounds concern uniform unknown-budget comparison at the critical supplied certificate, with exact counts and fixed boundary windows. They are not full-history impossibility results, every-budget logarithmic minimax penalties, claims about arbitrary interior statistics, or noisy-bit guarantees. The positive results include arbitrary starts; the lower witnesses are stationary. The constants are deliberately conservative. The word example is not empirical validation, and no learned-predictor experiment is asserted.

Attention, counting, and Markov representation. Uniform selected averaging and positional routing are standard operations in RASP [20]. Histogram constructions using uniform mixing also appear in Behrens et al. [2]. Edelman et al. [7, Proposition 2.1] give a two-layer bigram mechanism with previous-token copying; Rajaraman et al. [19] and Ekbote et al. [8] give shallow conditional k-gram representations under their respective positional and normalization models. These works preclude presenting shallow empirical Markov prediction as a new representation principle. The structural singleton masks used here are an explicitly paid different interface, not a bound for dense-causal-only attention. Their learning results and empirical studies address specified training settings; none is imported to claim that the present gate is learned. Lepage et al. [13] provide further empirical Markov-prediction context, comparing trained models with smoothed count estimates under different transition-matrix distributions.

Precision models. Merrill and Sabharwal [17, Proposition 1] analyze rounding individual attention probabilities. Our finite evaluator instead sums integer indicators exactly and rounds the aggregate once. These operations have different precision requirements: h describes the exported normalized bank, while addresses, accumulators, comparisons, and division use wider exact words. Balancing a statistical error with quantization error is standard estimation methodology; the assertion here is the matched upper and lower for this fixed bank. It is not an optimized-encoder or whole-network precision lower bound.

Earlier project results and scope of this consolidation. The prior finite-bank analysis [1], dated 2 November 2026, already proves real-error gate robustness and matching precision, numeric-noise, and source-edit rates for eight floored triple frequencies and four floored half-prefix frequencies, with a finite sequential evaluator. The present direct $J/C/D$ interface adds sequential clipping; its nearest-rounded bank has a different lower proof and a different canonical post-rounding fault port. The implementation makes local copying, Boolean/ReLU features, scalar-grid extraction, and hybrid shape explicit. These are interface and realization refinements of that earlier phenomenon. Separately rounded representations are not automatically equivalent after taking sums or differences, even when their unrounded counts determine each other.

8 Discussion

The main theorem identifies the price of a specified restriction on positional observations. Exact transition counts with short boundary windows cannot retain the known-budget scale for one map across all budgets at a fixed horizon. A proportional old-visit gate recovers the scale with a fixed positive fraction of the raw trajectory, or with a single computed bit after a whole-word scan. Its stability under approximate statistics makes a finite realization possible without exact recovery of the gate bit.

The complete argument relies on both sides of the information comparison: a gated reciprocal-occupation bound for the upper, and a legal histogram- and boundary-preserving transport for the

lower. Neither a generic robust selector nor a generic constrained-risk inequality removes those experiment-specific obligations. The external comparisons given here are not an exhaustive novelty certification.

A same-information classical counter already achieves the statistical guarantee with logarithmic auxiliary storage. The attention construction therefore supplies a transparent realization, not a speed, memory, or statistical advantage over that counter. Learning the positional decision in a specified trainable architecture, or imposing dense causal masks and bounded logits throughout, would require separate assumptions and evidence. No such result is asserted here.

The exact supplemental bit in Section 5 is fault-free after construction. Transmission through noisy channels is a separate problem, treated in the project's companion analyses of known channels, uncertain channels, and conditionally bounded noise; it is not covered by the numeric-bank replacement model of Section 6. In particular, a numeric coordinate perturbation, a bit error in an encoded message, and a corrupted arithmetic register are different experiments. The companion-work links and the claim-to-source ledger accompany this manuscript rather than adding unrelated coding theorems to its proof sequence.

A Finite-state estimates and upper bounds

Throughout, a departure count excludes the terminal state. For a chain $(Y_i)_{i=0}^{M}$, write

$$J_u = \sum_{i=0}^{M-1} \mathbf{1}\{Y_i = u\}, \qquad J = J_{Y_M}.$$

For the binary order-two experiment, $Y_i = (X_i, X_{i+1})$, $X_0, \dots, X_{M+1}$ contains M departures, and $q_u = \theta_u(d(u))$ with $d(a,b) = 1-a$. Let C_u count the corresponding designated successors and put $C = C_{Y_M}$ and $q = q_{Y_M}$. Primed counts are computed on the corrupted word. Every bound below permits an arbitrary initial law. The all-state certificate, when invoked, is $P^2(u,v) \geq \beta\Pi(v)$ for every pair of states, with $\beta > 0$. The final pair is protected, so clean and corrupted counts always select the same actual query row.

A.1 Universal occupation and clean prediction

Lemma A.1 (Actual-terminal occupation)**.** *For any homogeneous chain on d states, any initial law, any $M \geq 0$, and any integer $w \geq 1$,*

$$\Pr(J_{Y_M} < w) \leq \min\left\{1, \frac{d^2 w}{M+1}\right\}.$$

No irreducibility or stationary-start assumption is needed.

Proof. For each t set $J(t) = \sum_{i<t} \mathbf{1}\{Y_i = Y_t\}$, $b_t(v) = \Pr_v(J(t) < w)$, and $B_t = \max_v b_t(v)$. If $t \geq \ell$, the full count dominates the count in the final ℓ-step suffix. Conditioning on the suffix's initial state and using homogeneity gives $B_t \leq B_\ell$. On every path, each state can occur at most w times with fewer than w earlier occurrences. Hence $\sum_{t=0}^{M} \mathbf{1}\{J(t) < w\} \leq dw$. Consequently

$$(M+1)B_M \leq \sum_{t=0}^{M} B_t \leq \sum_{v} \sum_{t=0}^{M} b_t(v) \leq d^2 w.$$

Average over the initial law and use the probability cap. □

Lemma A.2 (Binary clean empirical comparison)**.** *For every homogeneous binary order-two source, every initial law, and $M \geq 64$,*

$$\mathbb{E}\left[\mathbf{1}\{J > 0\}\left|\frac{C}{J} - q\right|\right] \leq \frac{512}{\sqrt{M+1}}.$$

The expression inside the indicator is evaluated only when $J > 0$.

Proof. For each of the four rows u, generate an independent stack of $2M$ Bernoulli(q_u) variables $\xi_{u,j}$, independently of the initial pair. At the jth departure from u, use $\xi_{u,j}$ to choose whether the successor is $d(u)$. For every fixed finite path, the used stack prefixes have exactly the product of its transition probabilities; thus this realizes the original law, including boundary probabilities. Write $S_{u,j} = \sum_{i=1}^{j}(\xi_{u,i} - q_u)$. Pathwise, $C - Jq = S_{Y_M,J}$ on $J > 0$. The stacks are not asserted independent after conditioning on J, Y_M, or any selection event.

For a centered variable in an interval of length one, its log moment generating function has second derivative equal to a tilted variance, which is at most $1/4$. Integrating twice gives $\mathbb{E}e^{\lambda V} \leq e^{\lambda^2/8}$.

Thus $e^{\lambda S_{u,j}-j\lambda^2/8}$ is a nonnegative supermartingale. Stopping at its first level crossing before the deterministic time $2v$ gives, for $x > 0$,

$$\Pr\left(\max_{j\le 2v} S_{u,j} \ge x\right) \le e^{-\lambda x+2v\lambda^2/8} \le e^{-x^2/v}, \qquad \lambda = 2x/v.$$

The stopping inequality here is finite: the expectation of the stopped process is at most its initial value by summing conditional expected increments through time $2v$. Apply the same argument to $-S$ and take a union over rows. For $Z_v = \max_u \max_{j\le 2v} |S_{u,j}|$,

$$\Pr(Z_v \ge x) \le 8e^{-x^2/v}. \tag{A.1}$$

For dyadic integers $v \le M$, let $E_v = \{v \le J < 2v\}$ and $p_v = \min\{1, 32v/(M+1)\}$. Lemma A.1 gives $\Pr(E_v) \le p_v$. Without any independence assumption, tail integration and $a = \sqrt{v\log(8/p_v)}$ yield

$$\mathbb{E}[Z_v \mathbf{1}_{E_v}] \le \int_0^\infty \min\{p_v, 8e^{-x^2/v}\}\, dx \le p_v a + \frac{p_v v}{2a} \le 2p_v\sqrt{v\log(8/p_v)}.$$

For the middle inequality, bound the Gaussian tail above a by inserting $x/a \ge 1$ and integrating. This layer contributes at most $2p_v\sqrt{\log(8/p_v)}/\sqrt{v}$ to the desired risk. Put $N = M+1$. Summing layers with $v \ge N/32$ costs at most

$$\frac{2\sqrt{32\log 8}}{1-2^{-1/2}} N^{-1/2} < 56N^{-1/2}.$$

If v_0 is the largest dyadic integer below $N/32$, then $N/64 \le v_0 < N/32$ and $\log(N/(4v_0 2^{-j})) \le 3(j+1)$. The remaining layers cost at most

$$\frac{64\sqrt{3}}{\sqrt{32N}} \sum_{j\ge 0} 2^{-j/2}\sqrt{j+1} \le \frac{64\sqrt{3}}{\sqrt{32N}}(1-2^{-1/2})^{-2} < 236N^{-1/2}.$$

Only a numerical series was extended; all stack maxima use at most $2M$ variables. The layers partition $1 \le J \le M$, including the last layer when M is dyadic. The safe constant 512 follows. □

A.2 Return cycles and binary hitting scales

Lemma A.3 (Finite-state return identity)**.** *In a finite irreducible chain with stationary law p, the first positive return T to u has mean $\mathbb{E}_u T = 1/p_u$. If $H_u = \max_v \mathbb{E}_v D_u$, where D_u is the ordinary hitting time, zero at u, then $1/p_u \le 1 + H_u$.*

Proof. Irreducibility and finiteness give a fixed block length and a positive minimum probability of hitting u within a block, uniformly over starting states. Repeated conditioning gives geometric tails for hitting and return times, so the following expectations are finite. Let $\nu(v) = \mathbb{E}_u \sum_{j=0}^{T-1} \mathbf{1}\{Y_j = v\}$. First-step conditioning inside this sum gives

$$(\nu P)(v) = \mathbb{E}_u \sum_{j=1}^{T} \mathbf{1}\{Y_j = v\} = \nu(v),$$

since $Y_0 = Y_T = u$. A finite irreducible chain has a unique stationary probability: for completeness, if η is another one, the ratios $\eta(v)/p_v$ are harmonic for the reversed transition matrix; a maximal ratio propagates along every positive reversed edge and hence is constant. Thus $\nu = (\mathbb{E}_u T)p$. Exactly one visit to u occurs before T, so $\nu(u) = 1$, proving the identity. A positive return is one step followed by an ordinary hit, proving $\mathbb{E}_u T \le 1 + H_u$. □

Lemma A.4 (Support and weighted hitting scale)**.** *A binary order-two source with a positive all-state certificate has either full stationary support or a singleton absorbing diagonal support. In the full-support case, with $q = (a, b, e, \tau)$ and $\overline{H}_u = \max\{H_u, 1/\Pi(u)\}$,*

$$\sum_u \Pi(u) q_u \overline{H}_u \le \frac{40}{\sqrt{\beta}}. \tag{A.2}$$

Proof. Stationarity of consecutive pair marginals gives $\Pi = (y, x, x, w)$, and stationary flow into each diagonal gives

$$ya = xe, \qquad w\tau = xb. \tag{A.3}$$

If $x > 0$, the certificate entries $P^2(10, 01) = ea \ge \beta x$ and $P^2(01, 10) = b\tau \ge \beta x$ force $a, b, e, \tau > 0$. The flow identities then imply $y, w > 0$. If $x = 0$, the stationary law is on diagonals; each supported diagonal is absorbing by stationarity. Two absorbing supported diagonals contradict the positive certificate. This proves the classification. Summing a certificate row gives $\beta \le 1$. Full support and the certificate imply irreducibility.

Assume full support and put $\gamma = (a, 1, 1, \tau) \ge q$. The two certificate entries just used, together with (A.3), also give

$$e^2 \ge \beta y, \qquad b^2 \ge \beta w. \tag{A.4}$$

Here, for example, $ea \ge \beta x = \beta ya/e$ and $a, e > 0$. For target 00, write h_{01}, h_{10}, h_{11} for the mean hit times. The first-step equations are

$$h_{01} = 1 + (1 - b)h_{10} + bh_{11}, \quad h_{10} = 1 + (1 - e)h_{01}, \quad h_{11} = \tau^{-1} + h_{10}.$$

They give

$$h_{01} = \frac{b + 2\tau}{e\tau}, \quad h_{10} = h_{01} - \frac{b}{\tau} - 1, \quad h_{11} = h_{01} + \frac{1 - b}{\tau} - 1,$$

and therefore $H_{00} \le b/(e\tau) + 2/e + 1/\tau + 1$. For target 01, the three hit times from $00, 10, 11$ are $1/a$, $1 + e/a$, and $1/\tau + 1 + e/a$, so $H_{01} \le 2/a + 1/\tau + 2$. Exchanging symbols gives the other two bounds:

$$H_{10} \le 1/a + 2/\tau + 2, \qquad H_{11} \le e/(ba) + 2/b + 1/a + 1.$$

The substitution exchanges 00 with 11, 01 with 10, and (a, b, e, τ) with (τ, e, b, a); it does not assume $b = e$. Using (A.3) and (A.4),

$$\sqrt{\beta}\, yaH_{00} \le \sqrt{\beta}\,(w + 2x + ew/b + xe) \le \sqrt{\beta}\,(w + 3x) + \sqrt{w} \le 5,$$
$$\sqrt{\beta}\, xH_{01} \le 2\sqrt{\beta}\, y/e + \sqrt{\beta}\, w/b + 2\sqrt{\beta}\, x \le 2\sqrt{y} + \sqrt{w} + 2x \le 5.$$

The symbol-exchanged inequalities give the same bound for $10, 11$. Lemma A.3 implies $\overline{H}_u \le H_u + 1$; hence $\Pi(u)\gamma_u \overline{H}_u \le 6/\sqrt{\beta} \le 10/\sqrt{\beta}$. Sum over four states and use $q \le \gamma$. □

A.3 First hits, endpoint probabilities, and joint tails

Lemma A.5 (Target endpoint and occupation bounds)**.** *For any d-state chain, possibly reducible, and $v \ne u$,*

$$\Pr_v(D_u = t) \le \frac{d}{t + 1}, \qquad t \ge 1. \tag{A.5}$$

For an irreducible chain, fix a target with stationary mass $p > 0$, maximum ordinary hitting mean H, and $h = \max\{H, 1/p\}$. For every initial state i and every integer $\ell \geq 1$, with $J_u(\ell) = \sum_{j=0}^{\ell-1} \mathbf{1}\{Y_j = u\}$,

$$\Pr_i(Y_\ell = u) \leq 4dp(1 + h/\ell), \tag{A.6}$$

$$\Pr_i(J_u(\ell) < \ell p/16) \leq 2e^{-\ell/(8h)}, \tag{A.7}$$

$$\Pr_i(Y_\ell = u, J_u(\ell) > 4\ell p + 1) \leq 16dp(1 + h/\ell)e^{-\ell/(2h)}. \tag{A.8}$$

Proof. Reciprocal return coefficients and first hits. For any finite stochastic matrix P, let T_j be the jth positive return to u, allowing defective returns, and let $F(z) = \mathbb{E}_u[z^{T_1}; T_1 < \infty]$. Strong Markov at returns gives $\Pr_u(T_j = s) = [z^s]F(z)^j$. All strong-Markov uses here can be obtained by conditioning on each possible stopping time and state and multiplying the subsequent transition probabilities; no conditional independent-sample claim is involved. Let P_{-u} delete row and column u. Writing P in target/rest block form, direct summation over the excursion before the first return gives $F(z) = zP_{uu} + z^2 P_{u,-u}(I - zP_{-u})^{-1}P_{-u,u}$. The block determinant identity therefore yields

$$1 - F(z) = \frac{\det(I - zP)}{\det(I - zP_{-u})}.$$

These are formal series, so no convergence issue arises. The identity $-\log\det(I - zA) = \sum_{s\geq 1} \operatorname{tr}(A^s)z^s/s$ follows by differentiating the determinant, expanding $(I - zA)^{-1}$, and integrating from zero. Consequently

$$\sum_{j\geq 1} \frac{\Pr_u(T_j = s)}{j} = \frac{\operatorname{tr}(P^s) - \operatorname{tr}(P^s_{-u})}{s} \leq \frac{d}{s}, \qquad s \geq 1. \tag{A.9}$$

Its left side is nonnegative; the upper bound uses nonnegative entries of P^s_{-u} and diagonal entries of P^s at most one. Now change only row u to send u deterministically to $v \neq u$. In this auxiliary stochastic chain, the first positive return has law $1 + D_u$ under the original start v. Retain only the first-return term of (A.9) to prove (A.5). The auxiliary chain need not be an order-two source.

A target-specific endpoint bound. Return now to the irreducible chain. Let $g(v) = \mathbb{E}_v D_u$. First-step equations and Lemma A.3 give $(Pg - g)(v) = -1 + \mathbf{1}\{v = u\}/p$. Telescoping expectations shows

$$\mathbb{E}_v J_u(s) = p\{s + \mathbb{E}_v g(Y_s) - g(v)\} \leq p(s + H). \tag{A.10}$$

A loop-rotation identity is

$$sP^s(u, u) = \sum_v \mathbb{E}_v[J_u(s)\mathbf{1}\{Y_s = v\}] \leq dp(s + H).$$

Indeed, for each $j < s$, the corresponding sum is $\sum_v P^j(v, u)P^{s-j}(u, v) = P^s(u, u)$. For $v \neq u$, split the first-hit convolution at $t = \lfloor s/2 \rfloor$. For early hits the return factor is at most $dp(1 + 2H/s)$. For late hits (A.5) is at most $2d/s$, and (A.10) from u bounds the sum of return factors by $p(\lceil s/2 \rceil + H)$. The resulting bound is at most $dp(2 + 2/s + 4H/s) \leq 4dp(1 + H/s)$. It also holds for $v = u$; the early sum may be empty at $s = 1$. This proves the slightly stronger version of (A.6) with H instead of h.

Return moments and positive transforms. The family consisting of ordinary hitting times from nontarget states and the positive return from u is closed under conditioning on survival and the current state. Its means are at most h. For each member V and integer $r \geq 1$,

$$\mathbb{E}V^r \leq r!h^{r-1}\mathbb{E}V. \tag{A.11}$$

To prove this by induction, use $V^r \le r\sum_{j=0}^{V-1}(V-j)^{r-1}$, condition each summand on survival through time j, and apply the induction hypothesis and the uniform mean bound h to the residual law. The sum of the survival probabilities is $\mathbb{E}V$. With $\lambda = 1/(4h)$, expansion in nonnegative powers gives

$$\mathbb{E}_i e^{\lambda D_u} \le \frac{4}{3}, \qquad \mathbb{E}_u e^{\lambda T_1} \le e^{2\lambda/p}, \qquad f := \mathbb{E}_u e^{-\lambda T_1} \le e^{-3\lambda/(4p)}. \tag{A.12}$$

For the last inequality use $e^{-x} \le 1 - x + x^2/2$ and $\mathbb{E}T_1^2 \le 2h/p$; the first two follow from the geometric-series bound on the moments. If $K = \lceil \ell p/16 \rceil$, fewer than K visits in $0, \dots, \ell - 1$ means $D_u + T_1 + \cdots + T_{K-1} \ge \ell$. These are fresh independent return cycles after the first hit, before any horizon restriction. Exponential Markov and (A.12) bound this probability by $(4/3)e^{-\lambda\ell + 2\lambda(K-1)/p} \le (4/3)e^{-7\lambda\ell/8} \le 2e^{-\ell/(8h)}$. This includes $K = 1$ and proves (A.7).

A joint tail by an exact change of path measure. Let $h_i = \mathbb{E}_i e^{-\lambda D_u}$, with $h_u = 1$. The inequality $e^{-x} \ge 1 - x$ gives $h_i \ge 3/4$ and $f \ge 3/4$. Define

$$Q_{ij} = e^{-\lambda} P_{ij} h_j / h_i \quad (i \ne u), \qquad Q_{uj} = e^{-\lambda} P_{uj} h_j / f.$$

The hitting and return equations show that every row sums to one. This auxiliary matrix preserves directed support, so is irreducible. Its ordinary hitting law is the original law tilted by $e^{-\lambda D_u}/h_i$, whence $H_Q \le 2H$. Its positive return law is tilted by $e^{-\lambda T_1}/f$, and

$$\frac{1}{2p} \le \mathbb{E}_Q T_1 \le \frac{2}{p}.$$

For the lower bound, use $\mathbb{E}[T_1 e^{-\lambda T_1}] \ge 1/p - \lambda(2h/p) = 1/(2p)$ and $f \le 1$; the upper bound uses $f \ge 3/4$. Kac for Q gives $p_Q \le 2p$ and $\max(H_Q, 1/p_Q) \le 2h$. The endpoint bound already proved yields $Q^\ell(i,u) \le 16dp(1 + h/\ell)$. For every length-$\ell$ path from i to u, telescoping the factors gives

$$Q(\text{path}) = P(\text{path}) e^{-\lambda\ell} f^{-J_u(\ell)} / h_i.$$

The count is exactly the number of target departures. Thus on $J_u(\ell) > 4\ell p + 1$, summing the reverse likelihood relation gives

$$\Pr_i(Y_\ell = u, J_u(\ell) > 4\ell p + 1) \le h_i e^{\lambda\ell} f^{4\ell p} Q^\ell(i,u) \le 16dp(1 + h/\ell) e^{-\ell/(2h)}.$$

This proves the joint bound with its stationary-mass factor; no product of an unconditional upper tail and an endpoint probability was taken. □

A.4 Weighted imbalance, including unsupported starts

Lemma A.6 (Half-window imbalance)**.** *For $M \ge 64$, put $n = \lfloor M/2 \rfloor$ and $J_{1,u} = \sum_{i=0}^{n-1} \mathbf{1}\{Y_i = u\}$. Every binary source with a positive all-state certificate, under every initial law, satisfies*

$$\mathbb{E}\left[q_{Y_M} \mathbf{1}\{J > 0, J_{1,Y_M} < J/256\}\right] \le \frac{65536}{M\sqrt{\beta}}.$$

Proof. We first prove a general irreducible-chain statement. Let $m = M - n$, $J_{2,u} = \sum_{i=n}^{M-1} \mathbf{1}\{Y_i = u\}$, and assign arbitrary weights $0 \le q_u \le 1$. Write $W_u = p_u q_u \overline{H}_u$, where p_u is stationary mass. For every $0 < c \le 1/256$ we claim

$$\mathbb{E}\left[q_{Y_M} \mathbf{1}\{J_{1,Y_M} < cJ\}\right] \le \frac{288d}{M} \sum_u W_u. \tag{A.13}$$

The strict imbalance event is empty at $J = 0$. Fix u with $np_u \geq 1$. If $J_{1,u} \geq np_u/16$ and $J_{2,u} \leq 4mp_u + 1$, then $m \leq 2n$ gives $J_{1,u}/(J_{1,u} + J_{2,u}) \geq 1/145 > 1/256$. Thus imbalance requires a first-window lower tail or a second-window upper tail. For the first event, condition on the whole history through Y_n, use the uniform future endpoint bound (A.6), and then the first-window lower tail (A.7). For the second event condition only on Y_n and use the joint bound (A.8). Since $m \geq n$,

$$\Pr_i(Y_M = u,\ J_{1,u} < cJ_u) \leq 24dp_u(1 + \overline{H}_u/n)e^{-n/(8\overline{H}_u)}.$$

After multiplication by q_u, this is $(24dW_u/n)(x+1)e^{-x/8}$, where $x = n/\overline{H}_u$. Since the maximum of $(x+1)e^{-x/8}$ is $8e^{-7/8} < 4$, it is at most $96dW_u/n$. If $np_u < 1$, use only the endpoint bound at time M. As $W_u \geq q_u$ and $p_u q_u < q_u/n$,

$$q_u \Pr_i(Y_M = u) \leq 4d(p_u q_u + W_u/M) \leq 8dW_u/n.$$

Sum over u, average over initial states, and use $n \geq M/3$ to prove (A.13). For full-support binary sources, Lemma A.4 now gives $288 \cdot 4 \cdot 40/(M\sqrt{\beta}) = 46080/(M\sqrt{\beta})$.

It remains to treat singleton stationary support, without discarding unsupported actual initial states. Suppose 00 is absorbing, so $a = 0$, and denote the other states by $A = 01$, $B = 10$, $C = 11$. An original path ending in this active set has never hit 00. Reflect the killing edge $B \to 00$ to $B \to A$, producing the auxiliary three-state chain

$$R(A,B) = 1 - b, \quad R(A,C) = b, \quad R(B,A) = 1, \quad R(C,B) = \tau, \quad R(C,C) = 1 - \tau.$$

Every original active path has probability at most its reflected-path probability: the only changed factor is $1 - e \leq 1$ on $B \to A$. The endpoint and all active-state counts are unchanged. Thus any nonnegative functional of these paths is dominated without conditioning on survival. For $b, \tau > 0$ this chain is irreducible, with

$$Z = 2\tau + b, \quad p_A = p_B = \tau/Z, \quad p_C = b/Z, \qquad p_A b\overline{H}_A \leq 1, \quad p_C \tau \overline{H}_C \leq 1. \tag{A.14}$$

Indeed $H_A = 1 + 1/\tau$ and $H_C = 2/b$ by first-step equations; $\overline{H}_A \leq 2 + 1/\tau$ and $\overline{H}_C \leq 1 + 2/b$ give the last inequalities since $b, \tau \leq 1$. Apply (A.13) with weights $(b, 0, \tau)$: the A, C contribution is at most $288 \cdot 3 \cdot 2/M = 1728/M$. If $b = 0$ or $\tau = 0$, approximate these two parameters by positive ones in the reflected chain. At fixed finite horizon, path probabilities and the weights are continuous, and the bound is uniform, so the same conclusion follows by taking a limit. No minorization-preserving approximation of the original source is being asserted. The remaining target B has weight e, and for an active start

$$e \Pr(Y_M = B) = \Pr(D_{00} = M + 1) \leq \frac{4}{M+2}, \tag{A.15}$$

by (A.5): the sole entry edge to 00 is $B \to 00$. Drop its imbalance event. An initial 00 contributes zero; mixtures follow by linearity. The total is at most $1732/M$, below the stated bound because $\beta \leq 1$. Exchanging symbols handles absorbing 11 with the same fixed designated-successor rule. □

A.5 Reciprocal occupation with an old visit

Lemma A.7 (Old-visit reciprocal count)**.** *For any homogeneous d-state chain and initial law, let $M \geq 2$, $0 \leq m \leq \lfloor M/2 \rfloor$, and*

$$O_u = \sum_{i=0}^{M-m-1} \mathbf{1}\{Y_i = u\}, \qquad O = O_{Y_M}.$$

For every $c > 0$,

$$\mathbb{E}\frac{\mathbf{1}\{O \geq cJ\}}{J+1} \leq \frac{1}{M}\left(3d^2 + 2d^3 \sum_{s=m+1}^{\lfloor (M-1)/2 \rfloor} \frac{1}{s}\right)$$
$$\leq \frac{3d^2 + 2d^3}{M}\left(1 + \log\frac{M}{m+1}\right),$$

where an empty sum is zero. In particular, the last numerator constant is 176 *for the pair chain. The statement includes $J = 0$ and $m = 0$.*

Proof. Starting at u, the event $\{Y_s = u, J_u(s) = j\}$ is exactly $\{T_j = s\}$ for $j \geq 1$: the initial visit is a departure, while the terminal visit is not. Equation (A.9), which allowed defective returns, gives

$$K_u(s) := \mathbb{E}_u \frac{\mathbf{1}\{Y_s = u\}}{J_u(s)+1} = \sum_{j \geq 1} \frac{\Pr_u(T_j = s)}{j+1} \leq \frac{d}{s}.$$

Fix an initial state $v \neq u$, and put $a_v(t) = \Pr_v(D_u = t)$. The event $Y_M = u, J_u = 0$ means $D_u = M$, so its contribution is at most $d/(M+1) \leq d/M$ by (A.5); it is impossible if $v = u$. On $J_u > 0, O_u \geq cJ_u$, integrality forces $O_u \geq 1$, hence the first hit is at most $M - m - 1$. Discarding the rest of the gate and using strong Markov gives the exact identity

$$\mathbb{E}_v \frac{\mathbf{1}\{Y_M = u, O_u \geq 1\}}{J_u + 1} = \sum_{t=1}^{M-m-1} a_v(t) K_u(M-t).$$

There is no target departure before t, so the full count is exactly the post-hit departure count, with no additional unit shift. For $t \leq M/2$, use $K_u(M-t) \leq 2d/M$ and $\sum_t a_v(t) \leq 1$; the contribution is at most $2d/M$. This also covers $v = u$, whose expression is just $K_u(M)$. For $t > M/2$, use $a_v(t) \leq 2d/M$ and write $s = M - t$; the contribution is at most

$$\frac{2d^2}{M} \sum_{s=m+1}^{\lfloor (M-1)/2 \rfloor} \frac{1}{s}.$$

Together with the zero-count term this is a per-target bound. Average over the start and sum over d terminal targets to obtain the first inequality. Finally, $\sum_{s=a}^{b} 1/s \leq 1/a + \log(b/a) \leq 1 + \log(M/a)$ for $b \leq M$, with the empty case automatic, gives the stated logarithmic bound. □

A.6 Protected edits and the two positional gates

Lemma A.8 (Pathwise gate reduction)**.** *Let D_u be a count of row u over a fixed set of departure indices, and at the protected row put $D = D_{Y_M}$. On the corrupted word use*

$$\widehat{q} = \begin{cases} C'/J', & J' > 0 \text{ and } 512D' \geq J', \\ 0, & \text{otherwise.} \end{cases}$$

For every allowed corruption kernel and every initial law,

$$R \leq \mathbb{E}\left[\mathbf{1}\{J > 0\}|C/J - q|\right] + 48k\mathbb{E}\frac{\mathbf{1}\{D \geq J/1024\}}{J+1} + \mathbb{E}[q\mathbf{1}\{J > 0, D < J/256\}]$$
$$+ \Pr(J < 6156k + 1) + \Pr(J < 3078k + 1). \tag{A.16}$$

In particular this applies to a first-half count or to the old-prefix count reconstructed by subtracting a suffix count from the total.

Proof. One edited symbol affects at most three departure triples; each changed triple changes the histogram's ℓ^1 norm by at most two. Pair-count windows affect at most two departures. Taking a union over the edits, including adjacent edits and indices straddling a split, gives the safe bounds

$$|J' - J|,\ |C' - C|,\ |D' - D| \le \delta := 6k.$$

When the old count is computed by subtraction, it is still exactly a direct count on each word, so the same bound applies without adding two separate perturbation bounds. Endpoint protection keeps the selected row fixed. Let $G' = \{J' > 0, 512D' \ge J'\}$, $H = \{D \ge J/1024\}$, and $B = \{J > 0, D < J/256\}$. On $G' \cap H$ with $J > 0$,

$$|C'/J' - q| \le |C/J - q| + \frac{8\delta}{J+1}.$$

For $\delta = 0$ the ratios coincide. For $\delta > 0$ and $J \ge 2\delta$, $J' \ge J/2$ and ratio subtraction gives a difference at most $4\delta/J \le 8\delta/(J+1)$. For $J < 2\delta$, use the unit bound on the difference of two probabilities. On $G' \cap H^c$, $D + \delta \ge D' \ge (J - \delta)/512$ whereas $D < J/1024$, so $J < 1026\delta = 6156k$. On $(G')^c \cap B^c$ with $J > 0$, if $J' = 0$ then $J \le \delta$; otherwise $D' < J'/512$ and $D \ge J/256$ give $J/256 - \delta < (J + \delta)/512$, hence $J < 513\delta = 3078k$. If the gate rejects on B, its loss is exactly q. The two displayed integer-threshold tails also cover $J = 0$. These cases give (A.16) pathwise and then in expectation, with $8\delta = 48k$. At $k = 0$ the false-status cases for $J > 0$ are impossible. Equality in the gate is accepted throughout. Since the comparison is pathwise, arbitrary permitted attack randomization does not change it. □

Proposition A.9 (Finite suffix and half-gate upper bounds)**.** *Let $M \ge 64$, $2 \le r \le M + 2$, and set*

$$m = \min\{r - 2, \lfloor M/2 \rfloor\}, \quad N = M - m, \quad L = 1 + \log \frac{M}{m+1}.$$

From the histogram obtain J'_u, C'_u, and from the last $m+2$ symbols obtain $T'_u = \sum_{i=M-m}^{M-1} \mathbf{1}\{Y'_i = u\}$. Use the gate in Lemma A.8 with $D'_u = O'_u = J'_u - T'_u$. This k, β-blind estimator has all-start risk

$$R \le \min\left\{1, \frac{65536}{\sqrt{M}} + \frac{65536}{M\sqrt{\beta}} + (8448L + 147744)\frac{k}{M}\right\}. \tag{A.17}$$

Alternatively, use $D'_u = J'_{1,u}$, the count over the first $\lfloor M/2 \rfloor$ departures. Its all-start risk is

$$R \le \min\left\{1, \frac{65536}{\sqrt{M}} + \frac{65536}{M\sqrt{\beta}} + 2^{23}\frac{k}{M}\right\}. \tag{A.18}$$

The latter estimator factors exactly through the histogram, protected pair, and the one bit $\mathbf{1}\{512J'_{1,Y_M} \ge J'_{Y_M}\}$.

Proof. The last $m + 2$ raw symbols reveal exactly m departure pairs and the terminal pair. Thus O'_u is precisely the count over indices $0, \dots, N - 1$, even though their raw order is not observed. In particular $r = 2$ means $m = 0$ and $O' = J'$; counting $r - 1$ suffix departures would incorrectly include the terminal pair. Since $N \ge \lceil M/2 \rceil$, the clean O is at least J_1. Lemma A.6 therefore bounds the bias in (A.16) by $65536/(M\sqrt{\beta})$. Lemma A.7, with $d = 4, c = 1/1024$, bounds its reciprocal term by $48k \cdot 176L/M = 8448Lk/M$. Lemma A.1 bounds the two tails together by

$$\frac{16(6156 + 3078)k + 32}{M+1} = \frac{147744k + 32}{M+1}.$$

Lemma A.2 and the remaining $32/(M+1)$ are bounded by $65536/\sqrt{M}$. The unit loss cap proves (A.17).

For the half gate we require a different reciprocal estimate, valid for any four-state chain and start:

$$\mathbb{E}\frac{\mathbf{1}\{J_1 \geq J/1024\}}{J+1} \leq \frac{130944}{M}. \tag{A.19}$$

To prove it, fix u and condition on the entire first-half history, whose terminal state is v and whose count of u is j. Let $m_2 = M - \lfloor M/2 \rfloor$ be the remaining length. Balance implies $J_{2,u} \leq 1023j$. In the suffix, the fixed-terminal event $Y_M = u, J_{2,u} \leq 1023j$ is a subset of the actual-terminal low-occupation event. Lemma A.1 gives conditional probability at most $16(1023j+1)/(m_2+1)$. Since $(J_u+1)^{-1} \leq (j+1)^{-1}$, the conditional contribution is at most $16368/(m_2+1)$, including $j=0$. Sum over four targets and use $m_2 + 1 \geq M/2$ to prove (A.19). This argument uses a deterministic prefix, not endpoint-conditioned independent samples; Lemma A.1 is valid at every suffix length. Apply Lemmas A.8, A.6, and A.2 again. The total corruption coefficient is

$$48 \cdot 130944 + 16(6156 + 3078) = 6433056 < 2^{23},$$

proving (A.18). The stated bit is exactly the gate decision on every corrupted word. If $J' = 0$ its value is one, but the decoder's separate $J' > 0$ test still returns zero. Thus neither this boundary case nor threshold equality requires another message. All edits occur before the exact message is formed; the bit channel itself is not being corrupted.

At $\beta = 1/M$, (A.17) gives directly, for $g_M(k) = \max\{M^{-1/2}, k/M\}$,

$$R \leq (131072 + 156192L)g_M(k) \leq 287264Lg_M(k) < 2^{19}Lg_M(k).$$

For $r = \max\{2, \lceil \alpha M \rceil\}$ with $0 < \alpha \leq 1$, $m+1 = \min\{r-1, \lfloor M/2 \rfloor + 1\} \geq \alpha M/2$. Indeed $r - 1 \geq \alpha M/2$ follows by separating $\alpha M \leq 2$ from $\alpha M > 2$. Hence $L \leq 1 + \log(2/\alpha)$. Likewise (A.18) becomes $\min\{1, 2^{17}/\sqrt{M} + 2^{23}k/M\}$ at the critical certificate. □

A.7 A count-only oracle when the budget is supplied

Proposition A.10 (Critical known-budget upper)**.** *At supplied certificate $\beta = 1/M$, a predictor using only the exact triple histogram, the protected pair, and the public budget k satisfies, for every source and initial law,*

$$R \leq 191488 \max\{M^{-1/2}, k/M\}, \qquad M \geq 64.$$

For $k \geq 1$ its designated probability is

$$\widehat{q}_k = \frac{(C' - 6k)_+}{J' + 6k + 2};$$

for $k = 0$ use $(C'+1)/(J'+2)$. This budget-dependent estimator is used only for the oracle comparison, not for the unknown-budget results.

Proof. A weighted occupation estimate. For an irreducible d-state chain with target masses p_u, let $h_u = \max\{H_u, 1/p_u\}$, $0 \leq q_u \leq 1$, and $W_u = p_u q_u h_u$. For every $A > 0$ and every initial law we first prove

$$\mathbb{E}\min\left\{q_{Y_M}, \frac{A}{J+2}\right\} \leq 384d^2\frac{A}{M} + \frac{192d}{M}\sum_u W_u. \tag{A.20}$$

Fix a target u. If $h_u > M$, the endpoint bound gives $q_u \Pr(Y_M = u) \le 4dp_u q_u(1+h_u/M) \le 8dW_u/M$. If $h_u \le M$, on $J_u \ge Mp_u/48$ the minimum is at most $48A/(Mp_u)$, and $\Pr(Y_M = u) \le 8dp_u$; this costs at most $384dA/M$. For the complementary event, put $n = \lfloor M/2 \rfloor$ and $m_2 = M - n$. Since $n \ge M/3$, $J_u < Mp_u/48$ implies $J_{1,u} < np_u/16$. Conditioning on the complete prefix through Y_n, use the uniform endpoint bound on the remaining m_2 transitions, and then the unconditional lower tail on the prefix. The weighted contribution is at most

$$8dp_u q_u(1 + 2h_u/M)e^{-M/(24h_u)} = \frac{8dW_u}{M}(x+2)e^{-x/24} \le \frac{192dW_u}{M}, \qquad x = M/h_u.$$

The last step uses $\sup_{x\ge 0}(x+2)e^{-x/24} = 24e^{-11/12} < 24$. Summing these two cases over targets proves (A.20). For full-support binary sources, Lemma A.4 yields

$$\mathbb{E}\min\left\{q_{Y_M}, \frac{A}{J+2}\right\} \le 6144\frac{A}{M} + \frac{30720}{M\sqrt{\beta}}. \tag{A.21}$$

For singleton support 00, the 00 contribution vanishes. Use the reflected chain from the proof of Lemma A.6. On paths ending active, original path weights are bounded above by reflected path weights, with exactly the same terminal state and departure counts. Thus the nonnegative minimum functional for targets A, C, with weights $(b, 0, \tau)$, is dominated pathwise in the same sum over paths. Equation (A.14) and (A.20), with $d = 3$, bound this part by $3456A/M + 1152/M$. The original target B contributes at most $e\Pr(Y_M = B) \le 4/(M+2)$ by (A.15). An initial 00 contributes zero. For boundary b or τ, approximate only the reflected chain by positive parameters: there are finitely many length-M paths, and each summand is continuous in probabilities and weights, including its minimum. The uniform bound therefore passes to the limit. Symbol exchange handles singleton 11. Since $\beta \le 1$, the larger bound (A.21) covers both singleton cases and arbitrary initial mixtures as well.

A pathwise lower envelope. For $k \ge 1$ put $B_k = 6k$, $D = J + 2$, $D' = J' + B_k + 2$, and $c' = (C' - B_k)_+$. The protected-edit perturbation bounds give

$$D' \ge D, \quad 0 \le c' \le C, \quad 0 \le C - c' \le 2B_k, \quad 0 \le D' - D \le 2B_k.$$

Thus $0 \le \widehat{q}_k \le C/(J+2)$, and

$$\frac{C}{D} - \frac{c'}{D'} = \frac{C - c'}{D} + \frac{c'}{D'}\frac{D' - D}{D} \le \frac{4B_k}{D}.$$

For $J > 0$, putting $p = C/J$ gives

$$0 \le p - \widehat{q}_k \le \min\left\{p, \frac{4B_k + 2}{J+2}\right\}.$$

Since $x \mapsto \min\{x, c\}$ is 1-Lipschitz on nonnegative reals, with $A = 24k + 2$ the triangle inequality implies

$$|\widehat{q}_k - q| \le \mathbf{2}\{J > 0\}|C/J - q| + \min\left\{q, \frac{A}{J+2}\right\}. \tag{A.22}$$

When $J = 0$, $C = 0$ and $\widehat{q}_k = 0$; since $A \ge 2$, the last term equals q, so the inequality still holds without defining C/J. This is valid for every permitted edit realization. Using Lemma A.2, (A.21), $\beta = 1/M$, and $A \le 26k$ gives

$$R \le \frac{31744}{\sqrt{M}} + 159744\frac{k}{M} \le 191488\max\{M^{-1/2}, k/M\}, \qquad k \ge 1.$$

For $k = 0$ the word is unedited. On $J > 0$, the add-one probability is within $1/(J+2)$ of C/J; on $J = 0$ pay unit loss. Lemma A.1 gives $\Pr(J = 0) \le 16/(M+1)$. For completeness, integer-threshold layer cake gives

$$\mathbb{E}\frac{1}{J+2} \le \int_0^1 \min\left\{1, \frac{16}{(M+1)t}\right\} dt \le \frac{8}{\sqrt{M+1}}.$$

Indeed a nonempty event $J < t^{-1} - 2$ uses threshold $\lceil t^{-1} - 2 \rceil \le t^{-1}$; for the last inequality use $\min\{1, z\} \le \sqrt{z}$ and integrate $t^{-1/2}$. Together with Lemma A.2 this is at most $536/\sqrt{M}$, below the asserted constant. No positional message or raw-window input is used in either budget branch. □

B Observation separation and lower bounds

Throughout this section a word has symbols $X_0, \dots, X_{M+1}$, pair states $Y_i = (X_i, X_{i+1})$, and departures $0 \le i < M$. Write $J_{ab} = \sum_{i<M} \mathbf{1}\{Y_i = ab\}$, $J_{\text{cross}} = J_{01} + J_{10}$, and $N_{abc} = H_{abc}(X)$. Thus Y_M is a terminal pair, not a departure. All risks below include independent predictor randomness. The observation $\mathcal{T}_r$ consists of the eight triple counts and the actual first and last r symbols.

B.1 An exact occupation mixture

For $0 < a, t < 1$, let Q_a be the stationary law with pair transition matrix, in the order $00, 01, 10, 11$,

$$P_a = \begin{pmatrix} 1-a & a & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & t & 1-t \end{pmatrix}, \qquad \Pi_a = \frac{(t/a, 2t, 2t, 1)}{1 + t/a + 4t}.$$

In particular the stationary initial factor is $\Pi_a(11) = a/[t + (1+4t)a]$.

Lemma B.1 (Occupation-mixture domination)**.** *Put $F = \{Y_0 = 11, Y_M = 00\}$ and $U = N_{001}$. For every integer j, every integer $u_0 \ge 1$, every interval $I \subset (0,1)$, and every additional event G (possibly depending on a),*

$$\int_I Q_a(F, J_{00} = j, U \ge u_0, G)\, \frac{da}{a} \le \frac{t}{u_0}.$$

Proof. Delete every transition departing from 00, both loops and exits. For $J_{00} = j$, exactly $L = M - j$ transitions remain. Their last transition is $10 \to 00$. Before that transition the retained path is on $B = 01, C = 10, D = 11$. A retained $C \to 00$ followed by an erased exit $00 \to B$ is represented by a marked $C \to B$ of weight $1/2$; a direct $C \to B$ is unmarked and also has weight $1/2$. The erased exit's factor a is restored separately. All other retained transitions keep their probabilities. The marked transition matrix is therefore

$$K(z) = \begin{pmatrix} 0 & 1/2 & 1/2 \\ (1+z)/2 & 0 & 0 \\ 0 & t & 1-t \end{pmatrix}.$$

For $L \ge 1$, define

$$h_{L,u} = \tfrac{1}{2}[z^u](K(z)^{L-1})_{D,C},$$

and set $h_{L,u} = 0$ for $L \le 0$ or $u < 0$. The final transition $C \to 00$ has weight $1/2$ and is not marked. A skeleton with u marks has u completed visits to 00 and one terminal visit. Distributing its $j - u$

loops among these $u+1$ holding slots gives exactly $\binom{j}{u}$ weak compositions. Each composition and marked skeleton recover one word, and deletion recovers both uniquely. Consequently, for $0 \le u \le j$,

$$Q_a(F, J_{00} = j, U = u) = \Pi_a(11)\binom{j}{u} a^u (1-a)^{j-u} h_{M-j,u}. \tag{B.1}$$

Infeasible cases have zero mass. The terminal holding slot is present even though Y_M is not a departure; there is no initial 00 slot.

For $u \ge 1$, the beta integral gives $\binom{j}{u}\mathrm{B}(u, j-u+1) = 1/u$. Thus, if $A_{j,u}$ has the $\mathrm{Beta}(u, j-u+1)$ law,

$$\int_0^1 Q_a(F, J_{00} = j, U = u)\,\frac{da}{a} = \frac{h_{M-j,u}}{u}\mathbb{E}\Pi_{A_{j,u}}(11) \le \frac{h_{M-j,u}}{u}. \tag{B.2}$$

The expectation is part of the equality: stationarity has not been cancelled. For $u = 0$ the corresponding factor is instead $h_{M-j,0}\int_0^1 (1-a)^j/[t + (1+4t)a]\,da$, and no $1/u$ formula is used. This is why $u_0 > 0$ is required.

To sum over marks at the exact outside horizon, put $R = K(1)$. This matrix is stochastic. Started at D, let b_n, c_n, d_n be its probabilities of B, C, D. Then

$$c_0 = 0, \quad c_1 = t, \quad b_n = c_{n-1}\ (n \ge 1), \qquad c_n = \tfrac{1}{2}c_{n-2} + t d_{n-1}\ (n \ge 2).$$

Induction gives $c_n \le 2t$. Hence, for each $L \ge 1$,

$$\sum_{u \ge 0} h_{L,u} = \tfrac{1}{2}c_{L-1} \le t.$$

First drop G and enlarge I to $(0,1)$ in the nonnegative integral. Then sum (B.2) over $u \ge u_0$ to obtain

$$\int_I Q_a(F, J_{00} = j, U \ge u_0, G)\,\frac{da}{a} \le \sum_{u \ge u_0} \frac{h_{M-j,u}}{u} \le \frac{1}{u_0}\sum_{u \ge 0} h_{M-j,u} \le \frac{t}{u_0}.$$

No conditioning on a prescribed number of outside marks or replacement of the horizon $M - j$ is involved. □

B.2 Relocating loops while preserving sampled windows

Lemma B.2 (Interior loop relocation)**.** *Suppose $Y_0 = 11, Y_M = 00$, $2 \le r$, and $2r \le M + 2$. Put $l = r - 2$, $h = M - r + 2$. If the interior vertices $Y_l, \dots, Y_h$ contain a visit to 11 strictly before a visit to 00, there is a deterministic transformation C_r preserving the complete triple histogram and the actual first and last r symbols, such that, with $j = J_{00}$,*

$$d_H(C_r(X), 1^{M+2-j}0^j) \le 3J_{\mathrm{cross}} + 4r - 5. \tag{B.3}$$

If at least one of the two movable diagonal-loop types is absent, the same conclusion holds without the ordered-visit condition.

Proof. Freeze departures $i < l$ and $i \ge h$. These are $2r - 4$ edges and spell exactly the sampled boundary symbols. From the interior path remove only the 111 and 000 loops, in numbers A_1 and A_0, retaining every other edge in order. Let $S_0, \dots, S_L$ be its vertices. Deleting self-loops neither removes a distinct-state visit nor reverses its order. Choose the first retained 11 and the first subsequent retained 00; insert all A_1 loops at the former and all A_0 at the latter. The choice is

deterministic. Deleting or inserting a loop preserves its attachment pair. Reinserting exactly the removed number of edges preserves the absolute suffix attachment index h, so the frozen edge labels and all sampled boundary symbols are unchanged. Every triple label is retained or reinserted once, proving histogram preservation. Unfrozen boundary-crossing triples need not keep their individual positions.

Here is the required positional count argument. A binary word of length N with a distinguished block of A_1 ones before a distinguished block of A_0 zeros has o other ones and z other zeros. If $j = A_0 + q$, $q \geq 0$, then

$$d_H(\text{word}, 1^{N-j}0^j) \leq \max\{z + q,\, 2o + z - q\}. \tag{B.4}$$

Indeed, write z_- for zeros before the target cut. The exact mismatch count is $2z_- + j - (A_0 + z)$. If a distinguished one lies after the cut, all distinguished zeros lie after the cut and $z_- \leq z$, giving $z + q$. Otherwise every one after the cut is among the o other ones; writing o_+ for their number, the other exact identity is $2o_+ + (A_0 + z) - j$, at most $2o + z - q$. This proof does not presume any order among the remaining symbols.

Put $u = N_{001}$, $v = N_{011}$, $p = N_{010}$, $w = N_{101}$, and $d = J_{\text{cross}}$. Endpoint flow gives

$$N_{100} = u + 1, \qquad N_{110} = v + 1, \qquad d = p + v + u + 1 + w.$$

Let f_1, f_0 be the frozen $111,000$ loop counts, so $f_1 + f_0 \leq 2r - 4$. The initial pair contributes two other ones. Counting the appended symbol of every remaining edge gives exactly

$$\begin{aligned} o &= 2 + u + v + w + f_1 = d + 1 - p + f_1, \\ z &= p + u + v + 2 + f_0 = d + 1 - w + f_0, \\ q &= j - A_0 = u + f_0 \geq 0. \end{aligned}$$

Since $u \leq d - 1$ and $d \geq 1$,

$$\begin{aligned} z + q &= d + 1 - w + u + 2f_0 \leq 2d + 2f_0, \\ 2o + z - q &= 3d + 3 - 2p - w - u + 2f_1 \leq 3d + 3 + 2f_1. \end{aligned}$$

Each is at most $3d + 4r - 5$, proving (B.3). If only one movable loop type occurs, put its block at any retained visit to its state, which exists because the loops were deleted there. The same two-block argument treats the absent block as empty; if both are absent use the identity transformation.

Finally, j is the sampled word's actual departure count. The comparison word has instead

$$J_{00}(1^{M+2-j}0^j) = j - 2 \quad (2 \leq j \leq M). \tag{B.5}$$

For this range its final pair and the transformed word's final pair are both 00, so all their Hamming differences are in the editable first M coordinates. □

B.3 A finite matched-observation lower bound

Proposition B.3 (Finite critical lower bound)**.** *Let*

$$c = 2^{-16}, \quad A = 2^{12}, \quad t = \frac{ck}{M}, \quad u_0 = \lfloor ck/4 \rfloor, \quad T = \lfloor M/8 \rfloor, \quad L_- = \log \frac{1}{4At}.$$

Assume integer M, k, r satisfy

$$\begin{gathered} M \geq 64, \quad 2 \leq r \leq k/128, \quad k \leq M/8, \quad ck \geq 1024, \\ At < 1/4, \qquad \frac{M}{k^2} \leq \frac{Ac^2}{2^{20}}. \end{gathered} \tag{B.6}$$

For any fixed predictor based on $\mathcal{T}_r$, let S be its supremum clean risk over the stationary laws Q_a, $At \le a \le 1/4$. Every such source satisfies the all-state certificate with supplied $\beta = 1/M$. There is a stationary source with the same supplied certificate and one source-independent strict-k corruption kernel for which

$$R_{\mathrm{null}} \ge \frac{u_0}{8Mt}\left[\left(\frac{t}{16} - S\right) L_-\right]_+ . \tag{B.7}$$

The kernel protects the final pair on every input and every output.

Proof. We give the source, probability, and kernel calculations separately.

Source legality and all sixteen certificate entries. Throughout $t \le c/8$, $a \ge At \ge t$, and $a \le 1/4$. Set $D = 1 + t/a + 4t$. Then $1 \le D \le 1 + 1/A + c/2 < 4/3 < 2$. The flow equalities $\Pi_a(00)a = \Pi_a(10)/2$ and $\Pi_a(11)t = \Pi_a(01)/2$, together with $\Pi_a(01) = \Pi_a(10)$, verify stationarity directly. Multiplication gives

$$P_a^2 = \begin{pmatrix} (1-a)^2 & (1-a)a & a/2 & a/2 \\ 1/4 & 1/4 & t/2 & (1-t)/2 \\ (1-a)/2 & a/2 & 1/4 & 1/4 \\ t/2 & t/2 & (1-t)t & (1-t)^2 \end{pmatrix}. \tag{B.8}$$

Every displayed entry is at least $t/2$: use $a \ge t$ and $a, t \le 1/4$ for the products and all other entries are immediate. Since $t/2 = ck/(2M) \ge 1/M$, every starting and destination state obeys $P_a^2(u, v) \ge M^{-1}\Pi_a(v)$. Also $a \ge At \ge 2/M$, so $a - \beta \ge a/2$.

Common-atom covariance and coupling bounds. The four paths to 11 of length three, from the states in order, have probabilities

$$(1-a)a/2, \qquad (1-t)^2/2, \qquad a/4, \qquad (1-t)^3.$$

These arise respectively from paths $00, 00, 01, 11$; $01, 11, 11, 11$; $10, 00, 01, 11$; and $11, 11, 11, 11$. Each is at least $\lambda = a/4$, so $P_a^3(x, \cdot) \ge \lambda\delta_{11}$ for every x. Split each three-step transition into this common atom and a residual probability law. Shared independent Bernoulli trials of probability λ couple any two chains at the first common atom. Fill in each three-step block by its conditional bridge; this preserves the full one-step path law. After meeting, use identical transitions. Thus the contraction coefficient at lag h is bounded by

$$\alpha_h = (1 - \lambda)^{\lfloor h/3 \rfloor}.$$

For a stationary chain W, a departure-state-set count with stationary mass p, or a count of a specified directed edge with stationary probability p, satisfies

$$\mathrm{Var}(V) \le \frac{7Mp}{\lambda}. \tag{B.9}$$

For an edge indicator I_i and $h \ge 1$, condition on the state after edge i. The conditional probability of edge $i + h$ is a $[0, 1]$-valued function after $h - 1$ transitions, whose deviation from its stationary value is at most α_{h-1}. Therefore $|\mathrm{Cov}(I_i, I_{i+h})| \le p\alpha_{h-1}$. For a state-set indicator use $\alpha_h \le \alpha_{h-1}$ instead. Since $\sum_{h \ge 1} \alpha_{h-1} = 3/\lambda$, expansion of the variance gives $Mp(1 + 6/\lambda) \le 7Mp/\lambda$, as claimed.

Couple a chain Z started at 00 to stationary W by the same block construction. Its meeting boundary σ has

$$\mathbb{E}\sigma \le 3/\lambda = 12/a.$$

Every departure-state or edge count in the two paths differs by at most σ pathwise: after boundary σ the paths agree, and there are at most σ earlier departures. No independence of σ and the stationary counts is needed.

Stationary reversal and count corrections. Let $\mathcal{R}(ab) = ba$. The stationary triple identity

$$\Pi_a(ab)P_a(ab, bc) = \Pi_a(cb)P_a(cb, ba)$$

holds for every triple: the only nonpalindromic cases are $001, 100$ and $011, 110$, given by the two flow identities above. It implies $\mathcal{R}P_a^*\mathcal{R} = P_a$, where P_a^* is the stationary reverse pair kernel. Reversing the symbols of a stationary word thus preserves its law. In particular, with $F = \{Y_0 = 11, Y_M = 00\}$ and $E = \{Z_M = 11\}$,

$$Q_a(F) = \Pi_a(00)P_{00}(E),$$

and conditional on F, the reversed pair path $Z_i = \mathcal{R}Y_{M-i}$ has law $P_{00}(\cdot \mid E)$. For $M \geq 64$, $\lfloor M/3 \rfloor \geq M/4$, and therefore

$$\alpha_M \leq e^{-aM/16} \leq \frac{16}{aM} \leq \frac{16}{Ack} \leq \frac{1}{4}.$$

Since $\Pi_a(11) = 1/D \geq 3/4$, we have $P_{00}(E) \geq 1/2$.

Put $V(W) = N_{001}(W)$ and $x = ck$. Its stationary mean is $Mt/D \in [x/2, x]$. Consider the four conditions

$$\sigma \leq x/16, \quad |V(W) - Mt/D| \leq x/8, \quad J_{\text{cross}}(W) \leq k/32, \quad J_{00}(W) \leq M/16. \tag{B.10}$$

Their failure probabilities, respectively, are at most

$$\frac{192M}{Ac^2k^2}, \quad \frac{1792M}{Ac^2k^2}, \quad \frac{458752M}{Ak^2}, \quad \frac{16}{A}. \tag{B.11}$$

The first follows from $\mathbb{E}\sigma \leq 12/a$ and $a \geq Ack/M$. For the second, (B.9) with $p = t/D$ gives $\operatorname{Var} V \leq 28Mt/a$; divide by $(ck/8)^2$. For the third, the mean is $4Mt/D \leq 4ck \leq k/64$, and $\operatorname{Var} J_{\text{cross}} \leq 112Mt/a \leq 112M/A$; divide by $(k/64)^2$. For the last use $\Pi_a(00) = t/(aD) \leq 1/A$ and Markov's inequality. Conditioning the joint coupling on E costs at most two. Under (B.6), twice the sum in (B.11) is bounded by

$$2\left(\frac{1984}{2^{20}} + \frac{458752c^2}{2^{20}} + \frac{16}{A}\right) < \frac{1}{16}.$$

On F, the exact reversal identities are

$$\begin{aligned} J_{00}(Z) &= J_{00}(Y) + 1, \\ J_{\text{cross}}(Z) &= J_{\text{cross}}(Y), \\ N_{001}(Z) &= N_{100}(Y) = N_{001}(Y) + 1. \end{aligned} \tag{B.12}$$

The first compares arrival indices $1, \ldots, M$ with departure indices $0, \ldots, M-1$; the second loses no mixed endpoint; the third combines triple reversal with entry-minus-exit flow at 00. Thus (B.10), the pathwise coupling bound, and $x \geq 1024$ imply for the original path

$$\begin{aligned} U &\geq x/2 - x/8 - x/16 - 1 \geq x/4 \geq u_0, \\ J_{\text{cross}} &\leq k/32 + ck/16 \leq k/16, \\ 2 \leq u_0 \leq J_{00} &\leq M/16 + ck/16 \leq M/8. \end{aligned}$$

The integer upper count bound is $J_{00} \leq T$.

Ordered visits in the interior. Let $l = r - 2$ and $h = M - r + 2$. Since $r \le M/1024$, $h - l = M - 2r + 4 \ge M/2$. Write B for a visit to 11 before a later visit to 00 within $Y_l, \dots, Y_h$. Because $M - h = l$ and $M - l = h$, reversal sends this event precisely to a visit to 00 before a later visit to 11 within $Z_l, \dots, Z_h$.

The ordinary expected hitting times of 00 and 11, zero at their targets and listed in state order, are

$$h_{00} = (0,\, 4 + 1/t,\, 3 + 1/(2t),\, 3 + 3/(2t)),$$
$$h_{11} = (3 + 3/(2a),\, 3 + 1/(2a),\, 4 + 1/a,\, 0).$$

For completeness, the first vector follows by solving $h_{01} = 1 + (h_{10} + h_{11})/2$, $h_{10} = 1 + h_{01}/2$, and $h_{11} = 1 + t h_{10} + (1 - t)h_{11}$, with $h_{00} = 0$. For the second solve $h_{00} = 1 + (1 - a)h_{00} + a h_{01}$, $h_{01} = 1 + h_{10}/2$, and $h_{10} = 1 + (h_{00} + h_{01})/2$, with $h_{11} = 0$. On $a, t \le 1/4$ the two maxima are $3 + 3/(2t)$ and $3 + 3/(2a)$. From any state at time l, hitting 00 and then 11 therefore has expected additional time at most

$$6 + \frac{3}{2t} + \frac{3}{2a} \le \frac{6}{t}.$$

This uses the strong Markov property at the first hit: partition by its finite time and state, apply the usual Markov property on each event, and sum. Distinct targets make the ordering strict. Markov's inequality over $h - l \ge M/2$ bounds the unconditioned failure probability by $12/(tM)$. Dividing by $P_{00}(E) \ge 1/2$ gives

$$Q_a(B^c \mid F) \le \frac{24}{ck} \le \frac{3}{128}.$$

This estimate is uniform in the time-l state; it assumes no independence from the final state or the counts.

Let G be the intersection of B with $U \ge u_0$, $J_{\mathrm{cross}} \le k/16$, and $2 \le J_{00} \le T$. The preceding count and hitting bounds give $Q_a(G \mid F) \ge 1 - 1/16 - 3/128 > 1/2$. Retaining the rare endpoint mass,

$$Q_a(F \cap G) \ge \Pi_a(00)/4, \qquad (a - \beta) Q_a(F \cap G) \ge t/16. \tag{B.13}$$

Indeed $P_{00}(E) \ge 1/2$, $a - \beta \ge a/2$, and $a\Pi_a(00) = t/D \ge t/2$.

The null and its full word masses. Take a stationary first-order binary source with $T_*(1,0) = q_* = 1/(2M)$ and $T_*(0,1) = \beta = 1/M$. Its symbol masses are $\pi_*(0) = 1/3$, $\pi_*(1) = 2/3$. Its stationary pair law is $\Pi_*(s,z) = \pi_*(s)T_*(s,z)$, and for all starting and destination pair states,

$$\frac{P_*^2((u,v),(s,z))}{\Pi_*(s,z)} = \frac{T_*(v,s)}{\pi_*(s)}.$$

For $(v,s) = (0,0), (0,1), (1,0), (1,1)$ these four ratios are $3(1-\beta)$, $3\beta/2$, $3q_*$, $3(1-q_*)/2$, respectively. They are all at least $\beta + q_* \ge \beta$, checking every pair-state certificate entry. The target at terminal 00 is Bernoulli(β), not Bernoulli(q_*). For $2 \le j \le T$ put $x^{(j)} = 1^{M+2-j}0^j$. These disjoint words have full stationary masses

$$v_j = \frac{2}{3}(1 - q_*)^{M+1-j} q_* (1 - \beta)^{j-1} \ge \frac{1}{8M}. \tag{B.14}$$

Bernoulli's inequality gives the prefix factor at least $1/2$ and the suffix factor at least $7/8$, so in fact the product is at least $7/(48M) > 1/(8M)$. These masses include the initial symbol and every one of the $M + 1$ transitions.

A predeclared strict kernel with identity rejection. For $2 \le j \le T$, define a finite measure on words by

$$\nu_j(dy) = \int_{At}^{1/4} Q_a(dy; F, U \ge u_0, J_{00} = j)\, \frac{da}{a}, \qquad \mu_j = \nu_j(\text{all words}).$$

Lemma B.1 gives $\mu_j \le t/u_0$. For $\mu_j > 0$, normalization ν_j/μ_j is a probability law. These laws depend only on declared M, k, r and the auxiliary family, not the true source, its parameters, its initial law, or a predictor seed.

On input $x^{(j)}$ with $\mu_j > 0$, draw Y once from ν_j/μ_j. If B holds, form the deterministic $C_r(Y)$ of Lemma B.2. Accept it only if its actual Hamming distance from $x^{(j)}$ is at most k and its final pair equals that of $x^{(j)}$. Otherwise output the unchanged input. On zero-mass labels and all other input words, also output the unchanged input. This defines one source-independent kernel on all words, with a strict pointwise budget and protected final pair, not merely an average budget. It uses one draw, not repeated sampling until acceptance.

On G, Lemma B.2 bounds the distance by $3k/16 + 4r \le 3k/16 + k/32 < k$, so every check succeeds, and

$$\mathcal{T}_r(C_r(Y)) = \mathcal{T}_r(Y).$$

The same statistic identity holds on every transformed acceptance; it is not asserted on identity rejections. These are the sampled alternative's boundary windows, not necessarily the null input's windows. Input-window edits are allowed and included in the Hamming check. The suffix label obeys exactly $J_{00}(x^{(j)}) = j - 2$ and $J_{00}(Y) = j$.

Nonnegative-loss ordering. For a sampled word y, let $\ell_a(y)$ be the predictor's expected absolute error against a on $\mathcal{T}_r(y)$, and let $\ell_*(y)$ be its corresponding error against β. On F the clean actual terminal target is a; the triangle inequality, also after averaging independent decoder randomness, gives $\ell_*(y) \ge a - \beta - \ell_a(y)$. For $\mu_j > 0$, (B.14) gives

$$\frac{v_j}{\mu_j} \ge \rho := \frac{u_0}{8Mt}.$$

First discard nonnegative losses from other null inputs and rejected proposals. On the remaining nonnegative accepted losses, replace v_j/μ_j by ρ. Next restrict to G, where acceptance and statistic matching hold. Only then use the signed triangle bound:

$$\begin{aligned} R_{\text{null}} &\ge \rho \int_{At}^{1/4} \mathbb{E}_{Q_a}[\mathbf{1}_{F\cap G}\ell_*(Y)]\, \frac{da}{a} \\ &\ge \rho \int_{At}^{1/4} ((a-\beta)Q_a(F\cap G) - \mathbb{E}_{Q_a}[\mathbf{1}_{F\cap G}\ell_a(Y)])\, \frac{da}{a} \\ &\ge \rho \left(\frac{t}{16} - S\right) L_-. \end{aligned}$$

The restricted alternative loss is bounded by its unconditional clean risk, hence by S. Risk is nonnegative, giving (B.7). In particular, no lower mass multiplier has been placed in front of a possibly negative quantity, and no endpoint-conditioned loss has been substituted for the unconditional target risk. □

B.4 Consequences for widths and supplemental messages

Proposition B.4 (Width and fixed-fraction consequences)**.** *For a prescribed sequence $2 \le r_M \le M+2$, a finite uniform factor relative to $g_M(k) = \max\{M^{-1/2}, k/M\}$ exists if and only if $\liminf_{M\to\infty} r_M/M > 0$, for stationary or arbitrary starts. For each fixed $0 < \alpha \le 1/4096$ and $r_M = \max\{2, \lceil \alpha M \rceil\}$,*

$$2^{-26} \log \frac{1}{32\alpha} \le C_{\mathrm{stat}}(\alpha) \le C_{\mathrm{all}}(\alpha) \le 2^{19} \left(1 + \log \frac{2}{\alpha}\right).$$

With exact counts and protected terminal pair, optionally also the initial pair, a post-corruption full-word encoder needs at least two supplemental symbols to attain a finite uniform factor; two suffice when its message is exact and fault-free.

Proof. Suppose first that the liminf is zero. Select an increasing subsequence on which $r_M/M \to 0$, and on that subsequence put

$$b_M = \max\{r_M, \sqrt{M}\}, \qquad k_M = \lfloor \sqrt{M b_M} \rfloor.$$

Then $b_M = o(M)$, $k_M/M \to 0$, $k_M/b_M \to \infty$, $r_M/k_M \to 0$, and $k_M/\sqrt{M} \to \infty$. Moreover $M/k_M^2 \to 0$, so every condition in (B.6) holds eventually. Here $t_M = ck_M/M$, $\rho = u_0/(8Mt_M)$, $L_- \to \infty$, and $\rho \to 1/32$. If one k-blind family had a finite admissible factor C, its clean risk on every alternative at this same M and supplied certificate would satisfy $S_M \le C/\sqrt{M} = o(t_M)$. For example, eventually $\rho \ge 1/64$ and $t_M/16 - S_M \ge t_M/32$, so

$$\frac{R_{\mathrm{null}}}{g_M(k_M)} \ge \frac{c}{2048} L_- \longrightarrow \infty,$$

since $g_M(k_M) = k_M/M$ eventually. This contradicts C. The alternatives may depend on the tested budget, but the predictor and its clean guarantee are the same map at the same horizon. These stationary witnesses also rule out an all-start guarantee.

Conversely, if the liminf is positive, choose $\delta > 0$ and M_0 so $r_M \ge \delta M$ for $M \ge M_0$. Put $m_M = \min\{r_M - 2, \lfloor M/2 \rfloor\}$. Then $(m_M + 1)/M$ is bounded away from zero for all sufficiently large M. The suffix estimate of Proposition A.9 is at most $2^{19}[1 + \log(M/(m_M+1))]g_M(k)$ at $\beta = 1/M$, so gives one uniform all-start factor eventually. A finite initial set of horizons is harmless: any prediction has risk at most one and $g_M(k) \ge M^{-1/2}$. This proves the exact criterion without monotonicity of the widths.

Now fix $\alpha > 0$ in the stated interval before sending M to infinity. For any family with finite admissible stationary factor C, put $k = 128 r_M$ at sufficiently large horizons. Then

$$\frac{k}{M} \to 128\alpha \le \frac{1}{32}, \quad t \to 128c\alpha > 0, \quad At \to 8\alpha < \frac{1}{4},$$

all finite-domain conditions eventually hold, and

$$L_- \to \log \frac{1}{32\alpha}, \qquad \frac{u_0}{8Mt} \to \frac{1}{32}, \qquad S/t \le \frac{C}{t\sqrt{M}} \to 0.$$

For these budgets $k \ge \sqrt{M}$ eventually. Using the deliberately loose $g_M(k) \le 2k/M$ in (B.7), the positive limiting bracket gives

$$C \ge \liminf_{M\to\infty} \frac{R_{\mathrm{null}}}{g_M(k)} \ge \frac{1}{32}\frac{c}{32} \log \frac{1}{32\alpha} = 2^{-26} \log \frac{1}{32\alpha}.$$

This holds for every admissible finite factor, so taking the infimum does not require an attained optimum. The upper follows from Proposition A.9: $m_M + 1 \geq \alpha M/2$, including $r_M = 2$, hence $1 + \log(M/(m_M + 1)) \leq 1 + \log(2/\alpha)$. The stationary class is a subclass of the arbitrary-start class. These arguments take the horizon limit for each fixed fraction and factor; they neither exchange limits nor give an every-budget logarithmic risk lower bound.

Finally, a one-symbol supplemental alphabet is constant, so leaves only counts and the supplied endpoint pair or pairs. At $r_M = 2$ even both endpoint pairs are insufficient by the necessity just proved. The exact one-bit factorization in Proposition A.9 supplies two symbols and has all-start risk at most $(2^{17} + 2^{23})g_M(k)$. This concerns a fault-free message computed by scanning the whole already corrupted word, not one raw coordinate or a message subject to further corruption. □

B.5 Elementary same-information oracle lower floors

Lemma B.5 (Critical oracle lower floors)**.** *For every $M \geq 64$ and $0 \leq k \leq M$, the stationary minimax risk with supplied $\beta = 1/M$ is at least $g_M(k)/96$, even if the predictor knows k and observes the entire corrupted word. Consequently this lower bound holds for each restricted observation considered here and for the arbitrary-start minimax risk.*

Proof. We give separate clean and edit witnesses; taking their maximum is permitted in a minimax lower bound.

Clean floor. Put $N = M + 2$ and use stationary IID Bernoulli sources

$$p_\pm = \tfrac{1}{2} \pm \frac{1}{16\sqrt{N}}.$$

Their pair chains satisfy $P^2(u, v) = \Pi(v)$ for every row, hence the supplied certificate. The target is $p_\pm$ at every terminal pair. The elementary Bernoulli divergence bound

$$d(p\|q) \leq \frac{(p-q)^2}{q(1-q)}$$

follows by applying $\log x \leq x - 1$ to its two summands. With $\Delta = p_+ - p_- = 1/(8\sqrt{N})$, it gives one-letter divergence at most $1/(8N)$. Independence makes the full-word divergence at most $1/8$, and their total variation distance is at most $1/4$. For clarity, this last inequality follows without a statistical independence assertion about a selected row: for finite probability laws P, Q, group their masses on $A = \{x : P(x) \geq Q(x)\}$ and its complement. The log-sum inequality (Jensen's inequality for $x \log x$) gives $D(P\|Q) \geq d(P(A)\|Q(A))$. For fixed q, the second derivative in p of $d(p\|q)$ is $1/[p(1-p)] \geq 4$; its value and first derivative vanish at $p = q$. Thus $D(P\|Q) \geq 2(P(A) - Q(A))^2 = 2\,\mathrm{TV}(P, Q)^2$.

Testing by the nearer target makes each error incur prediction loss at least $\Delta/2$. Every possibly randomized test has average error at least $(1 - \mathrm{TV}(P, Q))/2$, since the difference of its acceptance probabilities is at most total variation. Thus the average prediction risk is at least

$$\frac{\Delta}{4}(1 - \mathrm{TV}(P, Q)) \geq \frac{3}{128\sqrt{N}} \geq \frac{1}{64\sqrt{M}}.$$

Identity corruption is allowed at every k, including for a known-budget predictor.

Edit floor. For $k \geq 1$, use the uniform prior on IID Bernoulli parameters $p \in [1/4, 3/4]$. Put $b = k + 1$. If the first M symbols contain n ones, retain a uniformly chosen set of $r = b\lfloor n/b \rfloor$ of those ones, turning the others to zero. Leave the final two symbols U unchanged. This single kernel is source-independent and changes $n - r \leq b - 1 = k$ editable bits on every outcome. For a specified retained set T of size r, conditional on n,

$$\Pr(T \text{ retained} \mid n) = \frac{\binom{M-r}{n-r}}{\binom{M}{n}\binom{n}{r}} = \frac{1}{\binom{M}{r}}.$$

Hence conditional on r, U the output arrangement is uniform and independent of p, even after mixing the possible n in a bucket. The full observation factors through at most

$$D = 4(\lfloor M/b \rfloor + 1)$$

labels (r, U) followed by a p-independent channel. The four protected pair possibilities are counted, not conditioned away.

Realize that channel using a seed independent of p and of the label, and likewise use an independent decoder seed. Freeze both seeds. There are at most D possible prediction values. For the uniform prior on an interval of length $\ell = 1/2$, the union of radius-v neighborhoods of those values has prior mass at most $2Dv/\ell$, regardless of how a label is randomly selected conditional on p. Outside that union every possible prediction has error greater than v. Integrating this probability bound gives Bayes absolute-error risk at least

$$\int_0^{\ell/(2D)} \left(1 - \frac{2Dv}{\ell}\right) dv = \frac{\ell}{4D}.$$

Averaging the independent seeds preserves it. Consequently some parameter in the prior support has risk at least

$$\frac{1}{32(\lfloor M/(k+1) \rfloor + 1)} \geq \frac{k+1}{32(M+k+1)} \geq \frac{k}{96M}.$$

IID stationarity and $P^2(u, v) = \Pi(v)$ again give the same source class and supplied certificate, with target p at every row. Combining the clean floor and edit floor yields $g_M(k)/96$, also at $k = 0$. A predictor with less information can only have a larger minimax risk, and allowing arbitrary starts contains the stationary experiment. □

C Simultaneous finite-horizon competitive factors

We prove Theorem 3.1 using the finite estimates proved in the preceding two sections. No limit in M or r is taken. Throughout this section $\ell = 1 + \log((M+2)/r) \geq 1$, the certificate is $1/M$, and Γ denotes the finite-horizon quantities of (2.7), not the sequence factors C.

Proof of Theorem 3.1. **Upper bound, including overlapping windows.** Set $m = \min\{r - 2, \lfloor M/2 \rfloor\}$ and $L = 1 + \log(M/(m+1))$. Since $r - 1 \geq r/2$ and $\lfloor M/2 \rfloor + 1 \geq M/2$,

$$\frac{M}{m+1} \leq \max\left\{\frac{2M}{r}, 2\right\} \leq \frac{2(M+2)}{r}.$$

The last step uses $r \leq M + 2$ and covers full or overlapping windows. Hence $L \leq \ell + \log 2 \leq 2\ell$. Proposition A.9 gives one counts-and-suffix map, independent of k, with all-start risk at most

$2^{19} L g_M(k) \le 2^{20} \ell g_M(k)$ for every k. Stationary experiments are a subclass of arbitrary-start experiments, so taking the infima over the same maps also gives $\Gamma_{\mathrm{stat}}(M,r) \le \Gamma_{\mathrm{all}}(M,r)$.

Lower bound and the small-logarithm case. Fix any allowable map A and write $G = \Gamma_{\mathrm{stat}}(A; M, r) \le \sqrt{M}$. Lemma B.5, at $k = 0$, implies $G \ge 1/96$ even if full history were available. If $\ell < 64$, then $1/96 \ge 2^{-30}\ell$, proving the desired bound. For $\ell \ge 64$, there is nothing to prove when $G \ge 2^{-30}\ell$. Suppose instead

$$G < 2^{-30}\ell. \tag{C.1}$$

As $r \ge 2$ and $M \ge 2$, $(M+2)/r \le M$. Thus $\log M \ge \ell - 1 \ge 63$, and $M > 2^{80}$ because $80 \log 2 < 56 < 63$. Choose the integer budget

$$k = \left\lceil \max\{128r, M^{3/4}\} \right\rceil. \tag{C.2}$$

This budget depends only on (M, r); the same map A is used at this budget and at zero.

Every finite lower-domain condition. Use the constants from Proposition B.3, writing $A_0 = 2^{12}$ for its numerical constant to distinguish it from the map:

$$c = 2^{-16}, \qquad t = ck/M, \qquad u_0 = \lfloor ck/4 \rfloor, \qquad L_- = \log \frac{1}{4A_0 t}.$$

The choice of k gives $r \le k/128$, $k \ge M^{3/4}$, and

$$\frac{M}{k^2} \le M^{-1/2} < 2^{-40} = \frac{A_0 c^2}{2^{20}}, \qquad ck \ge 2^{-16} M^{3/4} > 2^{44} \ge 1024.$$

Moreover

$$\frac{r}{M} = (1 + 2/M) e^{1-\ell} \le 2e^{-63} < 2^{-79}.$$

Since $k \le 128r + M^{3/4} + 1$, this implies

$$\frac{k}{M} < 2^{-72} + 2^{-20} + 2^{-80} < 2^{-19} < \frac{1}{8}.$$

In particular k is a legal budget, $k \le M/8$, and $A_0 t = k/(16M) < 1/128 < 1/4$. Together with the original integer $M \ge 64, r \ge 2$, these verify every condition in (B.6). The windows needed in that proof are nonoverlapping since $r \le k/128 \le M/1024$. Also $ck \ge 1024$ implies $u_0 \ge ck/8$, so

$$\rho := \frac{u_0}{8Mt} \ge \frac{1}{64}. \tag{C.3}$$

The same map's clean risk. Let S be the supremum of the clean risks on the stationary alternatives in Proposition B.3. All qualify at the same supplied certificate, so $S \le G/\sqrt{M}$. For $x \ge 0$, differentiation shows that $(1+x)e^{-x/4}$ is maximized at $x = 3$, with value $4e^{-3/4} < 4$. Thus $\ell \le 1 + \log M \le 4M^{1/4}$ and (C.1) gives the finite inequalities

$$S < 2^{-28} M^{-1/4} \le 2^{-21} M^{-1/4} \le t/32. \tag{C.4}$$

The last step uses $k \ge M^{3/4}$ and $c/32 = 2^{-21}$. No asymptotic small-order estimate or distinct clean estimator is used.

A logarithm uniform in both parameters. Since $4A_0 c = 1/4$, $L_- = \log(4M/k)$. Put $x = \max\{128r, M^{3/4}\} \ge 1$; then $k = \lceil x \rceil \le 2x$, and

$$L_- \ge \min\left\{ \log \frac{M}{64r},\ \log(2M^{1/4}) \right\}.$$

The first term equals $\ell - 1 - \log(1 + 2/M) - \log 64$ and is at least $\ell - 2 - \log 64 \geq \ell - 8 \geq \ell/8$, using $\log(1 + 2/M) \leq 1$, $\log 64 < 6$, and $\ell \geq 64$. The second term is at least $(\ell - 1)/4 \geq \ell/8$. Consequently $L_- \geq \ell/8$.

Common-kernel comparison and infimum. Proposition B.3 supplies a qualifying stationary null and one strict source-independent kernel, preserving the actual final pair, with

$$R_{\text{null}} \geq \rho \left[(t/16 - S)\, L_- \right]_+ .$$

Its full proof above includes source legality, mixture normalization, window matching, and pointwise budget checks. Here the bracket is positive by (C.4). Since $k \geq M^{3/4} \geq \sqrt{M}$, one has $g_M(k) = k/M$. Therefore

$$G \geq \frac{R_{\text{null}}}{g_M(k)} \geq \frac{(1/64)(t/32)(\ell/8)}{k/M} = 2^{-30}\ell,$$

contradicting (C.1). The lower bound thus holds for every map and passes to its infimum without requiring attainment. The witnesses are stationary and hence are also available in the all-start class. This completes the finite-horizon proof. □

For clarity about the sequence consequences, if r_M/M has positive liminf, then ℓ_M is bounded eventually, and finitely many earlier horizons have finite factor; the explicit suffix maps give a single uniform family. If that liminf is zero, ℓ_M diverges on a subsequence, so the finite lower forbids a uniform factor for any family. For fixed α, the finite bounds give the logarithmic order, while the original fixed-α proof in Proposition B.4 retains its stronger $2^{-26}\log(1/(32\alpha))$ lower constant. Neither argument claims a logarithmic risk lower at every individual budget.

D A direct-count approximation interface

This section changes the representation of the half-gate statistics, not its statistical mechanism. An earlier twelve-coordinate construction already established the real-error half-gate argument and the finite-bank precision, noise, and source-edit phenomenon [1]. Here the supplied coordinates directly approximate J, C, D, rather than reconstructing J from two nonnegative successor coordinates. Sequential clipping makes this changed interface legal. In particular, the result is not a claim that exact recovery of a discontinuous gate bit is necessary, nor a new precision principle.

D.1 Statement and clipping

For every row $u = (a, b)$, retain $d(u) = 1 - a$, and define the clean counts

$$J_u = \sum_{i=0}^{M-1} \mathbf{1}\{Y_i = u\}, \qquad C_u = \sum_{i=0}^{M-1} \mathbf{1}\{Y_i = u,\ X_{i+2} = d(u)\}, \tag{D.1}$$

$$D_u = \sum_{i=0}^{\lfloor M/2 \rfloor - 1} \mathbf{1}\{Y_i = u\}. \tag{D.2}$$

Thus D denotes a first-half departure count, always normalized by M when used as a frequency. Primed counts are evaluated on the corrupted word Z. At the actual protected row $u = Y_M$, omit the subscript and write $q = \theta_u(d(u))$. The protected pair makes the selected row the same for the clean and corrupted words.

For real x and $a \le b$, write $\mathrm{clip}(x;a,b) = \min\{b, \max\{a,x\}\}$. Suppose an observation-measurable computation supplies, for every row, real values $\widetilde{J}_u, \widetilde{C}_u, \widetilde{D}_u$. Its only accuracy premise is the simultaneous pathwise bound

$$\max_u \max\{|\widetilde{J}_u - J'_u|, |\widetilde{C}_u - C'_u|, |\widetilde{D}_u - D'_u|\} \le e, \qquad e \ge 0. \tag{D.3}$$

This must hold on every allowed word and for every outcome of the computation's independent random coins. The computation has no access to the clean word or hidden source parameters. Its errors may be dependent across coordinates and may depend on the observed word. The number e is a proved error bound, not an input required by the decoder. Clip sequentially, using the already clipped first coordinate:

$$\overline{J}_u = \mathrm{clip}(\widetilde{J}_u; 0, M), \qquad \overline{C}_u = \mathrm{clip}(\widetilde{C}_u; 0, \overline{J}_u), \qquad \overline{D}_u = \mathrm{clip}(\widetilde{D}_u; 0, \overline{J}_u). \tag{D.4}$$

The approximate half-gate returns designated probability

$$\widehat{q} = \begin{cases} \overline{C}/\overline{J}, & \overline{J} > 0 \text{ and } 512\overline{D} \ge \overline{J}, \\ 0, & \text{otherwise.} \end{cases} \tag{D.5}$$

Equality accepts. A final observation-measurable probability $\widehat{q}_{\mathrm{out}} \in [0,1]$ with $|\widehat{q}_{\mathrm{out}} - \widehat{q}| \le \xi$, $\xi \ge 0$, is also allowed. The complementary symbol receives probability $1 - \widehat{q}_{\mathrm{out}}$.

Theorem D.1 (Approximate direct half-gate statistics)**.** *For every integer $M \ge 64$, every integer $0 \le k \le M$, every homogeneous binary order-two source with an all-state certificate $P^2(u,v) \ge \beta\Pi(v)$, $\beta > 0$, every initial law, and every legal source-independent edit kernel in (2.2), the decoder (D.4)–(D.5), with the pathwise premise (D.3), has unsquared total-variation risk*

$$R \le \min\left\{1, \frac{65536}{\sqrt{M}} + \frac{65536}{M\sqrt{\beta}} + 1072176\frac{6k+e}{M} + \xi\right\}. \tag{D.6}$$

The same decoder works simultaneously for all these parameters without knowing k, β, or e. At the supplied certificate $\beta = 1/M$,

$$R \le \min\left\{1, \frac{131072}{\sqrt{M}} + 6433056\frac{k}{M} + 1072176\frac{e}{M} + \xi\right\}. \tag{D.7}$$

The edit kernel does not observe the predictor's random seed, and the result permits arbitrary independent randomization satisfying the pathwise accuracy premise.

Lemma D.2 (Clipping with a perturbed upper endpoint)**.** *Let $0 \le c \le j \le M$ and $|\widetilde{j} - j|, |\widetilde{c} - c| \le e$. Set $\overline{j} = \mathrm{clip}(\widetilde{j}; 0, M)$ and $\overline{c} = \mathrm{clip}(\widetilde{c}; 0, \overline{j})$. Then $|\overline{j} - j| \le e$ and $|\overline{c} - c| \le e$. The same statement holds with c replaced by any $d \in [0, j]$.*

Proof. Projection onto $[0, M]$ does not increase distance to j. For the second coordinate, $\overline{c} \le \max\{\widetilde{c}, 0\} \le c + e$. For its lower bound, if $\widetilde{c} < 0$, then $c \le e$, so $\overline{c} = 0 \ge c - e$. If $0 \le \widetilde{c} \le \overline{j}$, clipping retains $\widetilde{c} \ge c - e$. If $\widetilde{c} > \overline{j}$, then $\overline{c} = \overline{j} \ge j - e \ge c - e$. These cases exhaust the possibilities, including $\overline{j} = 0$; no clipped coordinate is required to be integer. □

D.2 Complete pathwise perturbation argument

One symbol edit affects at most three departure triples. Changing a triple has histogram ℓ_1 cost at most two, so each full-count coordinate changes by at most $6k$. Restricting to the first half does not increase this bound. Adjacent edits and triples crossing the split are covered by the union bound; their affected triples are not assumed disjoint. Lemma D.2 therefore gives, on every edit and computation outcome,

$$|\overline{J} - J|, |\overline{C} - C|, |\overline{D} - D| \le \Delta := 6k + e. \tag{D.8}$$

The perturbation radius Δ may be an arbitrarily small noninteger. Define the events

$$G = \{\overline{J} > 0,\ 512\overline{D} \ge \overline{J}\}, \qquad H = \{D \ge J/1024\}, \qquad B = \{J > 0,\ D < J/256\}.$$

Both C/J, where defined, and the accepted clipped ratio belong to $[0, 1]$.

On $G \cap H \cap \{J > 0\}$, we claim

$$\left|\frac{\overline{C}}{\overline{J}} - q\right| \le \left|\frac{C}{J} - q\right| + \frac{8\Delta}{J+1}. \tag{D.9}$$

If $\Delta = 0$, the ratios coincide. If $\Delta > 0$ and $J \ge 2\Delta$, then $\overline{J} \ge J/2$, and subtracting ratios gives

$$\left|\frac{\overline{C}}{\overline{J}} - \frac{C}{J}\right| \le \frac{|\overline{C} - C|}{\overline{J}} + \frac{C|J - \overline{J}|}{J\overline{J}} \le \frac{4\Delta}{J} \le \frac{8\Delta}{J+1},$$

where the last step uses the integer inequality $J \ge 1$. If $0 < J < 2\Delta$, then $\Delta > J/2$ and $8\Delta/(J+1) \ge 1$, so the unit bound on the ratio difference suffices. This proves (D.9) for all real $\Delta \ge 0$.

On $G \cap H^c$, acceptance and (D.8) imply

$$D + \Delta \ge \overline{D} \ge \frac{\overline{J}}{512} \ge \frac{J - \Delta}{512}, \qquad D < \frac{J}{1024}.$$

Combining them yields $J < 1026\Delta$. On $G^c \cap B^c \cap \{J > 0\}$, if $\overline{J} = 0$ then $J \le \Delta$. Otherwise the strict rejection inequality and $D \ge J/256$ give

$$\frac{J}{256} - \Delta \le \overline{D} < \frac{\overline{J}}{512} \le \frac{J + \Delta}{512},$$

and hence $J < 513\Delta$. For $\Delta = 0$, neither false-status case with $J > 0$ is possible. Rejection on B has loss exactly q.

The occupation lemma has an integer threshold, so introduce

$$w_1 = \max\{1, \lceil 1026\Delta \rceil\}, \qquad w_2 = \max\{1, \lceil 513\Delta \rceil\}. \tag{D.10}$$

They cover the strict false-status inequalities and $J = 0$. In the separate $\overline{J} = 0$, $J > 0$ case, $J \le \Delta$ forces $\Delta > 0$, and then $J < 513\Delta$ as required. For every $\Delta \ge 0$, $w_1 + w_2 \le 1539\Delta + 2$, including $\Delta = 0$. The preceding cases, bounding any remaining loss by one, prove the pointwise reduction

$$\begin{aligned} |\widehat{q} - q| \le \mathbf{1}\{J > 0\} \left|\frac{C}{J} - q\right| + 8\Delta \frac{\mathbf{1}_H}{J+1} + q\mathbf{1}_B \\ + \mathbf{1}\{J < w_1\} + \mathbf{1}\{J < w_2\}. \end{aligned} \tag{D.11}$$

The ratio C/J is evaluated only when $J > 0$. At $J = 0$ the tail indicators cover the loss without defining that ratio. All right-hand terms are nonnegative. In particular, no consistency assumption that the approximate coordinates are counts of a word is used, and no exact agreement with the corrupted word's gate bit is asserted.

D.3 Averaging and normalized precision

Proof of Theorem D.1. The four stochastic inputs have complete proofs in Section A. With $D = J_{1,Y_M}$, they are precisely

$$\begin{aligned}
\mathbb{P}(J < w) &\le \min\{1, 16w/(M+1)\} \quad \text{for integer } w \ge 1 && \text{(Lemma A.1)},\\
\mathbb{E}[\mathbf{1}\{J > 0\}\,|C/J - q|] &\le 512/\sqrt{M+1} && \text{(Lemma A.2)},\\
\mathbb{E}[q\mathbf{1}_B] &\le 65536/(M\sqrt{\beta}) && \text{(Lemma A.6)},\\
\mathbb{E}[\mathbf{1}_H/(J+1)] &\le 130944/M && \text{((A.19) in Proposition A.9)}.
\end{aligned}$$

In particular these inputs hold for arbitrary starts, including starts outside stationary support. They require neither endpoint-conditioned independent departures nor independence of the perturbation errors. Apply them to (D.11), then use (D.10):

$$\begin{aligned}
\mathbb{E}|\widehat{q} - q| &\le \frac{512}{\sqrt{M+1}} + \frac{65536}{M\sqrt{\beta}} + \frac{8 \cdot 130944\Delta}{M} + \frac{16(1539\Delta + 2)}{M+1}\\
&\le \frac{65536}{\sqrt{M}} + \frac{65536}{M\sqrt{\beta}} + \frac{1072176\Delta}{M}.
\end{aligned}$$

Here

$$8 \cdot 130944 + 16(1026 + 513) = 1072176, \qquad \frac{512}{\sqrt{M+1}} + \frac{32}{M+1} \le \frac{65536}{\sqrt{M}}.$$

The final output error contributes at most ξ by the triangle inequality. A legal binary probability has loss at most one. Substitute $\Delta = 6k + e$ to prove (D.6); then $\beta = 1/M$ proves (D.7). The proof is pointwise before averaging over all permissible randomizations, which proves the stated simultaneous quantifiers. □

For a nonnegative integer h, let $Q = 2^h$ and separately round each of the twelve normalized primed statistics $J'_u/M, C'_u/M, D'_u/M$ to its nearest point of $\{0, 1/Q, \dots, 1\}$, with any fixed tie rule. Multiplication by M gives approximations satisfying $e \le M/(2Q)$. Thus (D.7) gives the explicit rounded-bank upper bound

$$R \le \min\left\{1, \frac{131072}{\sqrt{M}} + 6433056\frac{k}{M} + \frac{536088}{Q} + \xi\right\}. \tag{D.12}$$

Consequently $h \ge \lceil \frac{1}{2}\log_2 M \rceil$ and $\xi \le M^{-1/2}$ suffice for a constant multiple of $g_M(k)$, uniformly in the unknown budget. If a separate output stage has error at most 2^{-b}, its contribution is simply $\xi = 2^{-b}$; the output precision b is distinct from the bank precision h and the raw-window width r.

For known M, a straightforward integer-numerator representation of this bank uses at most $12(h+1)$ bits, plus the two exact protected endpoint bits. The extra numerator bit permits the grid value one. This is a representation statement, not a streaming algorithm with that workspace: rounding after each update can accumulate error. Nor does it prove that all twelve coordinates are necessary. Raw input storage, exact prefix selection, addressing, accumulation, pair/triple feature construction, division, output precision, weights, depth, attention work, and training are separate costs. In particular, a classical counter has the same statistical interface; there is no neural statistical advantage in (D.12).

E Precision of the fixed nearest-rounded positional bank

The lower bound in this section concerns a particular observation, not an arbitrary encoder with the same number of bits. The earlier finite-bank result already established a twelve-coordinate precision floor together with noise and source-edit bounds [1]. Its floored triple/half-count coordinates differ from the nearest-rounded direct $J/C/D$ coordinates below. Before quantization the two representations are linearly interconvertible, but separate coordinatewise quantization does not preserve that equivalence. We therefore give a complete lower proof for this changed cut. Its common-cell and two-point argument also includes $h = 0$; it does not assert a new statistical/quantization transition.

E.1 The exact observation and result

Fix integers $M \geq 64$ and $h \geq 0$, and set $Q = 2^h$. Let $\mathcal{N}_Q$ round to the nearest point of $\{0, 1/Q, \ldots, 1\}$ using a fixed tie rule. Use the counts in (D.1)–(D.2), and form the twelve-coordinate bank

$$\mathcal{B}_{M,h}(X) = \big(\mathcal{N}_Q(J_u/M), \mathcal{N}_Q(C_u/M), \mathcal{N}_Q(D_u/M)\big)_{u\in\{00,01,10,11\}}. \tag{E.1}$$

The decoder observes exactly

$$\mathcal{O} = (\mathcal{B}_{M,h}(X), E), \qquad E = (X_M, X_{M+1}), \tag{E.2}$$

and source-independent public format metadata. The endpoint is exact. No raw history, scratch counts, other token banks, rounding residuals, or additional feature port crosses this cut.

Let $\mathcal{C}_M$ be the clean class of homogeneous binary order-two sources with arbitrary initial law admitting a stationary pair law Π with $P^2(u, v) \geq \Pi(v)/M$ for every u, v. Define

$$R^*(M, h) = \inf_A \sup_{(\theta,\mu)\in\mathcal{C}_M} \mathbb{E}\ \mathrm{TV}\big(A(\mathcal{O}), \theta_{Y_M}\big). \tag{E.3}$$

The infimum permits arbitrary computation, any binary probability output, and arbitrary independent decoder seeds. It is not restricted to the gate decoder or to a finite output grid. Public randomization independent of the source and sample may be included in these seeds; source-dependent advice or randomization correlated with hidden history is not part of (E.2).

Theorem E.1 (Fixed-bank precision)**.** *For every integer $M \geq 64$ and every integer $h \geq 0$,*

$$\frac{1}{1600000}\max\{M^{-1/2}, 2^{-h}\} \leq R^*(M, h) \leq \min\{1, 667160\max\{M^{-1/2}, 2^{-h}\}\}. \tag{E.4}$$

Both lower components already hold on the stationary IID Bernoulli subclass with parameter $p \in [1/8, 1/4]$. Consequently the same lower bounds hold for an adversarial experiment that allows zero edits, provided its decoder has no more information than the fixed cut (E.2)*.*

E.2 Two elementary probability inequalities

For completeness, we give the general concentration and testing inequalities needed below. They apply to finite input laws; randomized decoder seeds will be handled separately without restricting their support.

Lemma E.2 (Bounded differences)**.** *Let F be a real function of independent finite-valued coordinates, and suppose changing coordinate i alone changes F by at most $c_i \geq 0$. With $V = \sum_i c_i^2 > 0$, for every $t > 0$,*

$$\mathbb{P}(|F - \mathbb{E}F| \geq t) \leq 2\exp(-2t^2/V).$$

If $V = 0$, F is constant on the input support.

Proof. Reveal the independent coordinates in order and let Z_i be the increments of the Doob martingale of F. Conditional on any revealed prefix, Z_i has mean zero and lies in an interval of length at most c_i. Indeed, compare two values of coordinate i by integrating over the same independent remaining coordinates; the resulting conditional expectations differ by at most c_i.

For a centered random variable Z in an interval $[a, b]$ of length c, put $g(\lambda) = \log \mathbb{E}e^{\lambda Z}$. Its second derivative is the variance under the exponentially tilted law. For any law on $[a, b]$, the pointwise inequality $(Z - a)(b - Z) \geq 0$ implies

$$\mathrm{Var}(Z) \leq (\mathbb{E}Z - a)(b - \mathbb{E}Z) \leq (b - a)^2/4.$$

The same bound therefore holds under every tilt. Since $g(0) = g'(0) = 0$, integrating twice gives $g(\lambda) \leq \lambda^2 c^2/8$ for either sign of λ. Apply this bound conditionally to each Z_i and iterate conditional expectations to obtain

$$\mathbb{E}e^{\lambda(F - \mathbb{E}F)} \leq \exp\left(\lambda^2 \sum_i c_i^2/8\right).$$

Markov's inequality, optimized at $\lambda = 4t/V$, gives $\mathbb{P}(F - \mathbb{E}F \geq t) \leq e^{-2t^2/V}$. Apply the same argument to $-F$ and add the two tails. The zero-sensitivity case is immediate. □

Lemma E.3 (Pinsker's inequality for finite laws)**.** *For finite probability laws P, Q,*

$$\mathrm{TV}(P, Q) \leq \sqrt{\mathrm{KL}(P\|Q)/2}.$$

Moreover, relative entropy adds across independent product factors.

Proof. If P has positive mass where Q has zero mass, the divergence is infinite and the bound is immediate. Otherwise set $A = \{x : P(x) > Q(x)\}$, so $\mathrm{TV}(P, Q) = P(A) - Q(A)$. For a subset with P-mass $a > 0$ and Q-mass b, its contribution to relative entropy is a times an expectation of $\log(P(x)/Q(x))$ under normalized P weights. Jensen's inequality for the convex function $-\log$ bounds this below by $a \log(a/b)$: indeed the normalized expectation of $Q(x)/P(x)$ is at most b/a, with equality if Q places no mass on zeros of P inside the subset. Zero masses follow by continuity or infinite divergence. Apply this log-sum inequality to A and its complement:

$$\mathrm{KL}(P\|Q) \geq \mathrm{kl}(P(A)\|Q(A)).$$

For fixed $q \in (0, 1)$, the binary divergence $f(p) = \mathrm{kl}(p\|q)$ satisfies $f(q) = f'(q) = 0$ and $f''(p) = 1/[p(1 - p)] \geq 4$. Integrating twice from q to p, in either direction, gives $f(p) \geq 2(p - q)^2$. Boundary cases follow by continuity or infinite divergence. Thus $\mathrm{KL}(P\|Q) \geq 2\,\mathrm{TV}(P, Q)^2$. Finally, a product-density ratio is a product of the factor ratios; expanding its logarithm into a sum and taking expectations proves additivity for independent products. □

E.3 A common rounding cell for the complete mean vector

Consider stationary IID Bernoulli(p) input symbols, with $p \in I = [1/8, 1/4]$, and put $t = \lfloor M/2 \rfloor / M$. The means of the unrounded normalized coordinates, in row order $00, 01, 10, 11$, are

$$\mathbb{E}_p(J_u/M)_u = ((1-p)^2, p(1-p), p(1-p), p^2), \tag{E.5}$$

$$\mathbb{E}_p(C_u/M)_u = (p(1-p)^2, p^2(1-p), p(1-p)^2, p^2(1-p)), \tag{E.6}$$

$$\mathbb{E}_p(D_u/M)_u = t((1-p)^2, p(1-p), p(1-p), p^2). \tag{E.7}$$

Every summand has the displayed law, including the final triple ending at X_{M+1}. Overlapping pairs and triples need not be independent for these identities, which use only linearity of expectation. The half bank has $\lfloor M/2 \rfloor$ summands and denominator M. For $M \geq 64$, $t \geq 1/3$.

On I, the four J-coordinate derivative magnitudes are bounded below by $3/2, 1/2, 1/2, 1/4$, respectively. The derivatives of the two distinct C polynomials are

$$(1-p)(1-3p) \geq 3/16, \qquad p(2-3p) \geq 13/64.$$

The D derivatives are t times the corresponding J derivatives. Thus all twelve mean functions are strictly monotone on I, and every derivative has magnitude at least $d = 1/12$. Duplicated mean functions do not cause a problem, and none of the coordinates is constant.

The rounding thresholds are

$$\mathcal{T}_Q = \{(j + 1/2)/Q : 0 \leq j \leq Q - 1\}.$$

Each monotone coordinate mean crosses at most Q thresholds. Take the distinct crossing points in the interior of I and include the two endpoints. These partition I into at most $12Q + 1$ open intervals. One such interval (a_*, b_*) has length

$$L = b_* - a_* \geq \frac{1/8}{12Q + 1} \geq \frac{1}{104Q}. \tag{E.8}$$

Choose $p_0 = a_* + L/3$ and $p_1 = a_* + 2L/3$. Then

$$\delta_p := p_1 - p_0 = L/3 \geq \frac{1}{312Q}, \qquad \delta_p \leq \frac{1}{24}. \tag{E.9}$$

The mean vectors at both parameters round to one common bank $\mathbf{b}_*$. More precisely, for each coordinate mean f and $j \in \{0, 1\}$,

$$\operatorname{dist}(f(p_j), \mathcal{T}_Q) \geq dL/3 \geq \frac{1}{3744Q} =: \rho. \tag{E.10}$$

To justify this at the first and last partition intervals as well, no threshold lies strictly between the values $f(a_*)$ and $f(b_*)$. A threshold outside this value interval is separated from $f(p_j)$ by at least its distance to the corresponding endpoint value; integrating the derivative lower bound over the relevant subinterval gives at least $dL/3$. This also handles thresholds outside $f(I)$, without extending the derivative bound beyond I. Endpoint ties are harmless because p_0, p_1 are interior. The argument includes $h = 0$, when the sole threshold is $1/2$.

E.4 Concentration of the actual bank and its joint endpoint law

Let F be any one of the twelve unrounded normalized coordinates. Changing one raw bit affects at most three relevant departure triples, so changes the count by at most three. Pair and half-pair coordinates in fact require at most two. There are $N = M + 2$ independent raw bits, so the bounded-difference constants satisfy

$$\sum_i c_i^2 \le (M+2)(3/M)^2 \le 10/M, \qquad M \ge 64.$$

Lemma E.2 yields

$$\mathbb{P}_p(|F - \mathbb{E}_p F| \ge \rho) \le 2\exp(-M\rho^2/5).$$

If every coordinate is strictly within ρ of its mean, the actual rounded bank equals $\mathbf{b}_*$. The union bound and (E.10) give, separately at p_0 and p_1,

$$\mathbb{P}_{p_j}(\mathcal{B}_{M,h} \ne \mathbf{b}_*) \le \epsilon := 24\exp\left(-\frac{M}{5\cdot 3744^2 Q^2}\right). \tag{E.11}$$

This is concentration of the random bank, not a substitution of the rounded expectation for the expectation of a rounded variable. No independence between bank coordinates or count summands is needed.

Set $C_0 = 500000000$. If $M \ge C_0 Q^2$, then $\epsilon \le 1/16$. For an elementary numerical verification, $C_0 > 30 \cdot 3744^2$, and the nonnegative Taylor terms for e^6 through degree ten already exceed 384. Hence $24e^{-C_0/(5\cdot 3744^2)} < 24/384 = 1/16$.

Let P_j be the joint law of $(\mathcal{B}_{M,h}, E)$ at p_j, and let $\widetilde{P}_j$ be the law of $(\mathbf{b}_*, E)$ with the same endpoint marginal. Couple them by retaining exactly the same endpoint E and replacing only the bank. Equation (E.11) implies

$$\mathrm{TV}(P_j, \widetilde{P}_j) \le \epsilon.$$

Under IID sampling, the endpoint marginal is $\mathrm{Bern}(p_j)^{\otimes 2}$. Couple the two endpoint bits using common independent uniforms at the two parameters. A union bound gives $\mathrm{TV}(\widetilde{P}_0, \widetilde{P}_1) \le 2\delta_p$. Therefore, when $M \ge C_0 Q^2$,

$$\mathrm{TV}(P_0, P_1) \le 2\epsilon + 2\delta_p \le \frac{1}{8} + \frac{1}{12} = \frac{5}{24} < \frac{1}{2}. \tag{E.12}$$

The bank is not asserted independent of its endpoint: the final departures explicitly use the protected bits. Keeping the same endpoint in the first coupling is what validates this argument.

E.5 Randomized two-point loss and the full-history floor

For any decoder let A denote its probability assigned to symbol one. In the IID submodel the true probability is p at every actual endpoint, so total variation loss is exactly $|A - p|$. For two finite observation laws P_0, P_1, their common subprobability measure has mass $1 - \mathrm{TV}(P_0, P_1)$. Append the same independent seed law to both observations, allowing arbitrary seed spaces. This preserves total variation and gives a common subprobability product measure of the same mass. On every observation and seed, $|A - p_0| + |A - p_1| \ge |p_1 - p_0|$. Integrating this inequality against that common measure, and using nonnegativity to bound the two risks below, proves

$$\max_{j=0,1} \mathbb{E}_j|A - p_j| \ge \frac{|p_1 - p_0|}{2}(1 - \mathrm{TV}(P_0, P_1)). \tag{E.13}$$

This covers arbitrary independent randomized decoders and public source-independent randomization. Apply (E.12) and (E.9) to obtain

$$R^*(M,h) \geq \frac{\delta_p}{4} \geq \frac{1}{1248Q}, \qquad M \geq C_0 Q^2. \tag{E.14}$$

For the complementary regime, we prove a floor even when the decoder sees all $N = M + 2$ raw bits. Take $p_0 = 1/8$ and $p_1 = p_0 + \delta$ with $\delta = 1/(16\sqrt{N})$, so both belong to I. The inequality $\log x \leq x - 1$ gives the Bernoulli divergence bound

$$\begin{aligned} \mathrm{KL}(\mathrm{Bern}(p_0) \| \mathrm{Bern}(p_1)) &\leq p_0 \left(\frac{p_0}{p_1} - 1 \right) + (1 - p_0) \left(\frac{1 - p_0}{1 - p_1} - 1 \right) \\ &= \frac{\delta^2}{p_1(1 - p_1)} \leq \frac{64}{7} \delta^2. \end{aligned}$$

The last bound uses $p_1 \in I$, where $p_1(1 - p_1) \geq 7/64$. Product additivity in Lemma E.3 bounds the full-history divergence by $N(64/7)\delta^2 = 1/28$, and therefore its total variation by $\sqrt{1/56} < 1/4$. Passing to any function of the history cannot increase total variation: each event in the output space has a preimage event in the history space. Independent randomization also cannot increase it, by integration over the common seed law. Consequently (E.13) proves

$$R^*(M,h) \geq \frac{3\delta}{8} = \frac{3}{128\sqrt{M+2}} \geq \frac{1}{64\sqrt{M}}. \tag{E.15}$$

The last inequality is equivalent to $9M \geq 4(M + 2)$, valid here. These IID tables are legitimate all-state witnesses: put $\theta_u(1) = p$ for every row and initialize the pair with $\Pi = \mathrm{Bern}(p)^{\otimes 2}$. After two transitions, $P^2(u, v) = \Pi(v)$ for every initial pair u, so the required certificate $1/M$ holds on every row.

E.6 Combining the regimes and interpreting precision

Proof of Theorem E.1. For $M < C_0 Q^2$, (E.15) gives

$$R^*(M,h) > \frac{1}{64\sqrt{C_0}\, Q} \geq \frac{1}{1600000 Q},$$

since $\sqrt{C_0} < 25000$. For $M \geq C_0 Q^2$, (E.14) gives this same weaker lower bound. Together with (E.15), both of which hold for every decoder, this proves

$$R^*(M,h) \geq \frac{1}{1600000} \max\{M^{-1/2}, Q^{-1}\}.$$

The parameter pair used for the statistical floor need not equal the pair used for the precision floor. Each is a lower bound for every decoder's source supremum, so their maximum is a lower bound for the same minimax risk.

For the upper bound, apply (D.12) with $k = 0$ and $\xi = 0$ to exactly the bank in (E.1). Its sequential clipping and ratio gate are permitted arbitrary decoders, and

$$R^*(M,h) \leq \min\left\{1, \frac{131072}{\sqrt{M}} + \frac{536088}{Q}\right\} \leq \min\left\{1, 667160 \max\{M^{-1/2}, Q^{-1}\}\right\}.$$

This proves the complete finite-domain statement. □

For any fixed $K > 0$, attaining $R^*(M,h) \le K/\sqrt{M}$ requires

$$Q \ge \frac{\sqrt{M}}{1600000K}, \qquad h \ge \tfrac{1}{2}\log_2 M - \log_2(1600000K).$$

Conversely, $h \ge \lceil \frac{1}{2}\log_2 M \rceil$ suffices by (D.12); a separate finite output stage with error at most 2^{-b} adds just that error. The necessity statement is about fractional precision of this deterministic nearest-rounded $J/C/D$ bank. It is not a lower bound for arbitrary encoders using $12(h+1)$ bits, arbitrary positional summaries, neural architectures, or total arithmetic precision. It gives neither a noise lower bound nor a k/M lower bound for this changed bank. Accumulation precision, addresses, raw input storage, score precision, parameters, masks, and other program resources are not lower bounded here. Additional hidden history or counts would change the premise. The deliberately loose crossover C_0 is a proof device, not a practical sample-size threshold.

F A finite positional attention realization

This section realizes the approximate-statistic interface by an explicitly specified finite causal attention/Boolean–ReLU program. Its purpose is to verify the observation cut and its costs. A sequential counter on the same input is a stronger computational baseline, described in Section F.8. Uniform aggregation of indicators and positional copying are established operations [7, 20]. Moreover, the earlier finite-bank construction [1] already contains a paid uniform positional bank, a real-error half-gate analysis, and a finite sequential evaluator. The construction here specifies singleton-copy heads, direct $J/C/D$ rails, nearest rounding, a canonical post-rounding replacement cut, and an explicit Boolean–ReLU suffix with a separately stated hybrid shape. It does not establish a new statistical precision phenomenon or a learned mechanism.

F.1 The finite model and its guarantee

For this section only, let $M \ge 1$ and $h, b \ge 0$ be integers. The parameter b denotes *output* precision; r remains the raw-window width used elsewhere. Set

$$n = M+3, \qquad q = M+2, \qquad Q = 2^h, \qquad \ell_a = \lceil \log_2(n+1) \rceil, \qquad p = \ell_a + b + h + 32. \tag{F.1}$$

The address width ℓ_a is distinct from the logarithmic competitive factor ℓ in Theorem 3.1. The input consists of the $M+2$ binary symbols $Z_0, \dots, Z_{M+1}$ and one synthetic zero token at position q. No one-hot representation, endpoint label, local pattern feature, or count is supplied without construction.

For $u = (a,c) \in \{0,1\}^2$, define on an arbitrary binary word

$$\begin{aligned} J'_u &= \sum_{i=0}^{M-1} \mathbf{1}\{(Z_i, Z_{i+1}) = u\}, \\ C'_u &= \sum_{i=0}^{M-1} \mathbf{1}\{(Z_i, Z_{i+1}) = u,\ Z_{i+2} = 1-a\}, \\ D'_u &= \sum_{i=0}^{\lfloor M/2 \rfloor - 1} \mathbf{1}\{(Z_i, Z_{i+1}) = u\}. \end{aligned} \tag{F.2}$$

An empty sum is zero. A *finite attention head* in the present model has explicit public structural masks and exact bounded-word evaluation of its allowed-key sum, followed by a specified rounding

operation. The integer work of that evaluator is charged separately from the number of attention layers. Pointwise maps consist of affine maps and ReLUs, with exact finitely represented coefficients and explicit live-wire holds. This is a hybrid computational model, not unit-cost real arithmetic or a finite-precision hardware softmax claim.

Theorem F.1 (Finite realization of the positional bank)**.** *For each M, h, b above there is a fixed causal finite program, depending only on these public parameters, with the following properties.*

1. *It has two attention layers and three heads in total, with at most two heads in one layer, query/key dimension one, maximum value dimension twelve, and score cap zero. Shared pointwise Boolean–ReLU maps have depth $O(p^2)$ and width $O(p)$ per token. The integer attention-evaluator work is additional and is charged below.*

2. *On every binary input word, it acquires the exact pair (Z_M, Z_{M+1}) and produces twelve master numerators $B_{u,A} \in \{0, \dots, Q\}$, $A \in \{J, C, D\}$, satisfying*

$$\left|\frac{B_{u,J}}{Q} - \frac{J'_u}{M}\right|, \quad \left|\frac{B_{u,C}}{Q} - \frac{C'_u}{M}\right|, \quad \left|\frac{B_{u,D}}{Q} - \frac{D'_u}{M}\right| \;\le\; 2^{-h-1}. \tag{F.3}$$

   The inequalities are simultaneous and deterministic.

3. *At the specified cut, the only data-dependent downstream inputs are those twelve numerators and the exact endpoint pair. Replace all twelve masters jointly by arbitrary canonical integers $B^*_{u,A} \in \{0, \dots, Q\}$ satisfying*

$$\max_{u,A} \frac{|B^*_{u,A} - B_{u,A}|}{Q} \le \delta. \tag{F.4}$$

   The resulting normalized coordinates have error at most $\eta = 2^{-h-1} + \delta$ from (F.2) divided by M. Replacements may be correlated and may depend on the observed word. Endpoint bits, metadata, comparators, and subsequent arithmetic remain exact; this is the entire numeric fault port.

4. *The readout selects the actual endpoint row, clips its C, D coordinates to $[0, J]$, and applies the gate $J > 0$, $512D \ge J$. Its output is a binary probability distribution with denominator 2^b, within 2^{-b} total variation of the corresponding exact rational gate output, including empty rows, gate equality, $h = 0$, and $b = 0$.*

5. *In the experiment of Theorem D.1, with $M \ge 64$ and supplied certificate $1/M$, its actual-terminal risk obeys*

$$R \le \min\left\{1, \frac{131072}{\sqrt{M}} + 6433056\frac{k}{M} + 1072176(2^{-h-1} + \delta) + 2^{-b}\right\}. \tag{F.5}$$

   The program is not supplied k or δ.

A complete sparse-list/sparse-circuit hybrid realization has conservative online bitwork $O(np^4)$ and total online peak storage $O(p^4 + n\ell_a + np^2)$ bits. A complete dense-mask/dense-circuit alternative has online bitwork $O(n^2p^2 + np^6)$ and total peak storage $O(p^5 + n^2 + n\ell_a + np^2)$ bits. These are alternative full representations, not interchangeable entries in one cost calculation. Section F.7 gives construction, input, addressing, mask, hold, cleanup, and output costs and a separately charged all-ReLU unrolling.

The endpoint in this theorem is computed from the input, not appended as an additional observation. The fault restriction in (F.4) is substantive: off-grid analog errors, corrupted endpoint bits, accumulator errors, and comparator faults are not covered. The construction and the following proof do not assume that the rounded or replaced bank is the count table of any binary word.

F.2 Causal masks and the twelve direct rails

Write $\rho(x) = \max\{x, 0\}$. Initially

$$\mathrm{raw}_t = Z_t \quad (0 \le t \le M+1), \qquad \mathrm{raw}_q = 0.$$

Public position information provides the query-type flag $U_t = \mathbf{1}\{t = q\}$ and the half flag

$$F_t = \mathbf{1}\{2 \le t < 2 + \lfloor M/2 \rfloor\}. \tag{F.6}$$

These flags are generated by public integer comparisons. Positions and flags cost $O(n\ell_a)$ stored bits and their generation is charged below.

First layer: two singleton copies. Both heads have $Q_t = K_t = 0$ and the one-dimensional value raw_t. Their permitted-key sets are

$$\mathcal{M}_t^{(1)} = \begin{cases} \{t-1\}, & t \ge 1, \\ \{t\}, & t = 0, \end{cases} \qquad \mathcal{M}_t^{(2)} = \begin{cases} \{t-2\}, & t \ge 2, \\ \{t\}, & t < 2. \end{cases} \tag{F.7}$$

Each row is nonempty and causal. Its only normalized attention weight is exactly one, so the returned bits, denoted $l_{1,t}$ and $l_{2,t}$, are exact copies. Raw bits and both copied bits are retained through paid identity holds. In particular,

$$(l_{2,i+2}, l_{1,i+2}, \mathrm{raw}_{i+2}) = (Z_i, Z_{i+1}, Z_{i+2}) \quad (0 \le i < M), \qquad (l_{2,q}, l_{1,q}) = (Z_M, Z_{M+1}). \tag{F.8}$$

The fallback copies at positions zero and one never enter the query's statistic. These masks are exact structural exclusions. They are not approximations obtained by assigning large scores in a dense causal row.

Local pointwise features. For each $u = (a, c)$ in lexicographic order, form

$$\begin{aligned} R_{t,u} &= (1 - (l_{2,t} \text{ XOR } a)) \text{ AND } (1 - (l_{1,t} \text{ XOR } c)), \\ V_{t,u,J} = R_{t,u}, &\qquad V_{t,u,C} = R_{t,u} \text{ AND } (\mathrm{raw}_t \text{ XOR} a), \qquad V_{t,u,D} = R_{t,u} \text{ AND } F_t. \end{aligned} \tag{F.9}$$

These are twelve binary rails, not an eight-successor histogram with an additional reconstructed row sum. On the legal key $t = i + 2$, substitution in (F.8) gives respectively

$$\mathbf{1}\{(Z_i, Z_{i+1}) = u\}, \qquad \mathbf{1}\{(Z_i, Z_{i+1}) = u,\ Z_{i+2} = 1-a\}, \qquad \mathbf{1}\{(Z_i, Z_{i+1}) = u,\ i < \lfloor M/2 \rfloor\}. \tag{F.10}$$

Thus the C rail uses precisely the designated successor. Each feature uses only its own token's local copies and the public half flag. In particular, no endpoint-conditioned query is broadcast to historical tokens.

Second layer: one uniform head. This head again has one-dimensional $Q = K = 0$, and its value is the twelve-dimensional vector in (F.9). Its mask is

$$\mathcal{M}_q^{(3)} = \{2, 3, \ldots, M+1\}, \qquad \mathcal{M}_t^{(3)} = \{t\} \quad (t \ne q). \tag{F.11}$$

The query has exactly M allowed keys. The key $M+1$ includes the last legal departure triple, whereas the synthetic query q is excluded. The terminal pair is not an additional departure. Every other row uses a causal singleton self mask; all those rows are evaluated and their outputs are

subsequently cleared. This choice is part of the construction and is not a dense causal-prefix restriction.

All allowed scores equal zero, so $\exp(0) = 1$ and the ideal normalized head is a uniform average. There is no approximation of a logarithm, exponential, anchor inversion, or content match. For each row with L allowed keys, sum each binary rail into an *exact* integer $S \in \{0, \dots, L\}$ and return its nearest grid numerator

$$\mathcal{R}_{h,L}(S) = \left\lfloor \frac{2QS + L}{2L} \right\rfloor. \tag{F.12}$$

The denominator L is public, obtained from the enumerated mask, and is never zero. The floor in this formula will be implemented by the finite division circuit below; it is not a primitive real-number operation. At q, summing (F.10) proves the all-word identity

$$(S_{q,u,J}, S_{q,u,C}, S_{q,u,D}) = (J'_u, C'_u, D'_u). \tag{F.13}$$

Equation (F.12) rounds QS/L to its nearest integer, with ties upward. It maps 0 to 0 and L to Q, and

$$\left| \frac{\mathcal{R}_{h,L}(S)}{Q} - \frac{S}{L} \right| \le \frac{1}{2Q}.$$

Together with (F.13), this proves (F.3) for every binary word, including $M = 1$ and $h = 0$. Counts are accumulated before this single rounding. Repeated h-grid rounding during accumulation would be a different algorithm.

F.3 Exact Boolean and ReLU arithmetic

All words use little-endian bit rails, with enough leading zeros for the stated operation. For bits x, y, the identities

$$\begin{aligned} x \text{ AND } y &= \rho(x + y - 1), & \text{NOT}(x) &= \rho(1 - x), \\ x \text{ XOR } y &= \rho(x - y) + \rho(y - x), & x \text{ OR } y &= x + y - \rho(x + y - 1) \end{aligned} \tag{F.14}$$

are exact. Each expression is a constant-size affine/ReLU circuit. A coordinatewise multiplexer is

$$\text{mux}(s, x, y) = (s \text{ AND } x) \text{ OR } ((\text{NOT}(s)) \text{ AND } y). \tag{F.15}$$

It equals x when $s = 1$ and y when $s = 0$. Thus no data-dependent selection below is free dynamic addressing.

Addition, subtraction, and comparison. For input bits x_i, y_i and carry c_i, a full adder has

$$v_i = x_i \text{ XOR } y_i, \qquad s_i = v_i \text{ XOR } c_i, \qquad c_{i+1} = (x_i \text{ AND } y_i) \text{ OR } (v_i \text{ AND } c_i).$$

Induction on i gives ordinary binary addition. A ripple chain on w-bit words uses $O(w)$ depth and active gates, with $O(w)$ live rails. Retain the final carry separately. Subtract y from x by adding $x + \text{NOT}(y) + 1$. Since this integer is $x - y + 2^w$, its final carry is one exactly when $x \ge y$, and its low w bits are $x - y$ modulo 2^w. Hence the same chain implements unsigned comparison and subtraction. Public shifts are explicit fixed wires with zero padding; their storage and holds are charged.

Lemma F.2 (Finite restoring division)**.** *For a positive w-bit denominator and a w-bit dividend, exact quotient and remainder can be computed with w fixed public iterations, $O(w^2)$ active Boolean/ReLU gates and depth, and $O(w)$ live rails. A layered representation that holds all live rails at every step has at most $O(w^3)$ gates and identity holds.*

Proof. Initialize remainder $R = 0$. Scan the dividend's bits from most significant to least. Given the next bit x_i, first form $\widetilde{R} = 2R + x_i$. Subtract the denominator d using the preceding circuit. Its final carry c says whether $\widetilde{R} \geq d$. Emit c as the next quotient bit, and replace the remainder by $\mathrm{mux}(c, \widetilde{R} - d, \widetilde{R})$.

After scanning any prefix P of the dividend, the invariant is $P = Q_P d + R$ with $0 \leq R < d$, where Q_P is the emitted quotient prefix. For the next bit, $2R + x_i < 2d$, so one subtraction suffices, proving the invariant inductively. Also $R \leq P$, and therefore $\widetilde{R} \leq 2P + x_i$, the next scanned dividend prefix. That prefix fits in the original w bits. Thus the shifted remainder fits without an uncharged extra magnitude bit. The subtraction's carry is one explicit additional rail. At the final iteration the invariant is the desired Euclidean division identity. Each of w iterations costs $O(w)$ active gates and depth. Holding $O(w)$ live rails through $O(w^2)$ layers gives the stated conservative $O(w^3)$ bound. □

The same sequence is a total Boolean circuit even on denominator zero; its arbitrary zero-denominator result will never be used. The readout below instead selects a strictly positive denominator before invoking this circuit.

Word bounds for the attention evaluator. A feature sum needs $\lceil \log_2(M+1) \rceil$ bits; h bits alone do not suffice. The rounding dividend and divisor satisfy

$$2QS + L \leq 2^{h+1}M + M < 2^{h+\ell_a+2}, \qquad 2L \leq 2M < 2^{\ell_a+1}. \tag{F.16}$$

The same bounds cover singleton rows because $M \geq 1$. Use $w_{\mathrm{att}} = \ell_a + h + 16$ bits, together with explicitly retained carries. The accumulation, the $(h+1)$-place shift, addition of L, and restoring division all fit. In particular, the accumulation is exact, and the rounding numerator is computed without overflow. This implements (F.12) entirely by the preceding finite gadgets.

F.4 Packed and scalar versions of the bank

The bank has two possible physical representations, with the same declared observation. In a packed version, each master is stored as its $h+1$ numerator bits. Reading these existing bits and padding them with zeros is paid fixed wiring. The high bit is necessary to represent $B = Q$, that is, the normalized value one.

In a scalar version, the head returns one exact grid scalar $H = B/Q$ per rail. The head's conversion from numerator bits to this scalar is the paid dyadic sum

$$H = \sum_{j=0}^{h} 2^{j-h} B_j.$$

It uses $O(h+1)$ terms with $O(h+1)$-bit coefficients. A scalar is *not* treated as exposing its bits at no cost.

Lemma F.3 (Paid extraction from a canonical scalar grid)**.** *For $H \in \{0, 1/Q, \ldots, 1\}$, the numerator bits of QH are extracted exactly by a ReLU circuit of depth and active-gate count $O(h+1)$, with $O((h+1)^2)$ conservative holds and $O(h+1)$-bit coefficients and scalar words. No floor on arbitrary real inputs is used.*

Proof. Initialize $A = QH$ and, in the fixed order $j = h, h-1, \ldots, 0$, set

$$B_j = \rho(A - (2^j - 1)) - \rho(A - 2^j), \qquad A \leftarrow A - 2^j B_j. \tag{F.17}$$

Initially A is an integer in $[0, Q]$. At the beginning of step j it is an integer in $[0, 2^{j+1} - 1]$. If $A < 2^j$, both ReLUs in (F.17) vanish. If $A \geq 2^j$, their difference is exactly one. The update leaves an integer in $[0, 2^j - 1]$. Induction gives all the bits and final remainder zero. At $H = 1$, the top bit B_h is one and all remaining bits are zero. At $h = 0$, the single step extracts either zero or one. Each step has constant active cost; retaining all emitted bits through later steps costs $O((h+1)^2)$ identity holds. Thresholds, scaling coefficients, and exact scalar values have $O(h+1)$-bit descriptions. □

For the scalar realization the replacement supplies a canonical scalar $H^* = B^*/Q$, and extraction occurs *after* replacement. Consequently it reveals only the replaced master, not its earlier value. On off-grid inputs the recurrence need not produce bits; such inputs are outside the fault model. Twelve parallel extractions, their scaling, their scalar head decodes, and their holds fit the p-based resource bounds below. Both representations therefore reach the same padded numerator rails by paid operations.

F.5 Endpoint selection and a total finite readout

Let $(e_0, e_1) = (Z_M, Z_{M+1})$ be the two copied endpoint bits. For each $u = (a, c)$ form the selector

$$s_u = (1 - (e_0 \text{ XOR } a)) \text{ AND } (1 - (e_1 \text{ XOR } c)).$$

Exactly one selector equals one. For each of the three coordinate types, start a zero word and traverse the four rows in fixed order, replacing it by $\text{mux}(s_u, B^*_{u,A}, \text{current word})$. The resulting words are the selected canonical numerators. This is a constant number of word muxes; it does not use an uncharged lookup indexed by the endpoint.

Denote the selected words by J, C_0, D_0, and set

$$C = \min\{C_0, J\}, \qquad D = \min\{D_0, J\}, \qquad G = \mathbf{1}\{J \geq 1\} \text{ AND } \mathbf{1}\{512D \geq J\}. \tag{F.18}$$

Each minimum is implemented by comparison and a mux. Lower clipping is automatic from unsigned canonical inputs. Multiply D by 512 with a nine-place shift and compare exactly. Gate equality therefore accepts, whereas $J = 0$ rejects, irrespective of the second comparison.

Choose the safe denominator and quotient by

$$J_{\text{safe}} = \text{mux}(G, J, 1), \qquad v = \left\lfloor \frac{2^b C}{J_{\text{safe}}} \right\rfloor, \qquad m = \text{mux}(G, v, 0). \tag{F.19}$$

The divisor is always positive. The shift and restoring division implement the displayed floor. Even on a rejected row the pre-mux quotient is at most 2^{h+b}. On acceptance, $0 \leq C \leq J$ gives $0 \leq m \leq 2^b$. On rejection $m = 0$. Compute the complement numerator $2^b - m$ by exact subtraction and route the designated probability using the endpoint bit:

$$(P_0, P_1) = \begin{cases} (2^b - m, m), & e_0 = 0, \\ (m, 2^b - m), & e_0 = 1. \end{cases} \tag{F.20}$$

Again this is a pair of bitwise muxes. The actual output is $(P_0/2^b, P_1/2^b)$, always in the binary simplex. A dyadic decoder $\sum_{i=0}^{b} 2^{i-b}(P_a)_i$ uses paid coefficients of $O(b+1)$ bits.

On acceptance, flooring gives $0 \leq C/J - m/2^b < 2^{-b}$; on rejection the rational and finite outputs both put zero on the designated symbol. Binary total variation is the absolute difference of that probability. This proves the output error bound, including $b = 0$, without a gate-margin assumption.

All readout arithmetic can use $w_{\text{out}} = h + b + 16$ bits. Indeed, $512D \le 2^{h+9}$, $2^b C \le 2^{h+b}$, the pre-mux quotient is no larger than this dividend, and the final probability numerators are at most 2^b. Constants, intermediate carries, and padded words are bounded by $2p$ rails. Together with (F.16), these bounds prove the absence of overflow throughout the construction.

The pointwise feature and suffix maps are shared and executed at *all* n tokens. The public flag U_t clears packed bank and endpoint inputs at nonquery tokens using muxes. For a scalar bank rail $H_t \in [0,1]$, the paid gate $\rho(H_t + U_t - 1)$ retains H_t at the query and is zero elsewhere. Thus nonquery banks are cleared before the cut without extracting the query's bits early. Scalar extraction after replacement is executed at all tokens, with zero inputs away from the query. The shared suffix still computes on those zero inputs; its work is not charged only at q. Only the query's output is retained.

F.6 Isolation of the replacement cut and statistical composition

The master cut is after the twelve query coordinates have been rounded and before endpoint-conditioned selection, comparison, division, or any readout fanout. Every downstream bank copy originates from the replaced master. In the scalar realization it also precedes the extraction in Lemma F.3. The endpoint bits and public h, b, type flags, and layout remain exact. Nonquery banks and endpoint rails are cleared by the public query flag.

Before the cut is made available to the suffix, raw-input arrays, lag-copy arrays, local-feature arrays, attention accumulators, rounding dividends and divisors, intermediate quotients and remainders, and any clean-master duplicates are released or overwritten. The suffix has no later attention layer and no raw-memory read. Registers retaining M or layout contain only public, data-independent values; they do not encode any discarded count. An equivalent interface signature is simply

$$\big((B^*_{u,A})_{u,A},\ (e_0, e_1),\ h, b\big) \longmapsto (P_0, P_1).$$

There is no original-bank handle in this signature. All initialization, clearing, and identity holds are included in the resource accounts.

This is a logical access restriction on the program, not a claim of secure physical erasure against a hostile host retaining its own copies. The external fault injector is not counted as free neural hardware: its construction and runtime are outside the nominal program. The theorem charges the program on either side of the stated replacement port, not an implementation of the adversary.

Completion of the proof of Theorem F.1. The copied-bit identity (F.8), the feature identity (F.10), and the exact mask (F.11) prove the twelve all-word sums (F.13). Exact finite division implements their nearest rounding, proving (F.3). The replacement triangle inequality yields

$$\left| \frac{B^*_{u,A}}{Q} - \frac{A'_u}{M} \right| \le \frac{1}{2Q} + \delta = \eta \qquad (A \in \{J, C, D\}).$$

For analysis only, multiply the normalized replacement coordinates by M, obtaining approximate counts with simultaneous error at most $e = M\eta$. No such multiplication by M is needed or supplied to the actual readout. The canonical J already lies in $[0, Q]$, and (F.18), after the common positive scaling M/Q, is exactly the sequential clipping and gate of Theorem D.1. Both the ratio and the comparison are unchanged by this scaling. Inconsistent replacement coordinates are permitted because that theorem does not require count realizability.

The protected endpoint equals the actual terminal pair and has been acquired by the first layer. Equations (F.19)– (F.20) give output error at most $\xi = 2^{-b}$. Applying Theorem D.1 with $e = M\eta$

and that ξ proves (F.5). This is the only stochastic step. The all-word interface identities need neither a source law nor an attack independence assumption. The statistical composition retains the source, strict-edit, and observation-measurability assumptions of the approximate gate theorem. The finite circuits above give the claimed shape; the next subsection verifies their full cost bounds. □

In particular, if $h, b \geq \lceil \frac{1}{2} \log_2 M \rceil$ and $\delta = O(M^{-1/2})$, (F.5) is $O(M^{-1/2} + k/M)$. Theorem E.1 calibrates the clean, fault-free *fixed bank* obtained at this cut. It does not make the present word widths or total program precision necessary. The phrase "half-log precision" can refer only to the fractional precision h of these normalized coordinates, not to all arithmetic, weights, addresses, input storage, or total memory.

F.7 Complete resource accounts

All following bounds are constructive asymptotic upper bounds from the gate recipes and fixed public schedules, with absolute constants. They are not timing measurements or assertions that a conventional softmax kernel executes the integer evaluator at unit cost. We count a bit operation at the elementary Boolean/arithmetic level and charge the selected program representation, addresses, and all holds. Only one representation is selected at a time.

Hybrid shape and coefficient descriptions. There are two attention layers: two one-bit copying heads followed by one twelve-value uniform head. All scores are zero; query/key dimensions are one. A constant number of $O(p)$-word selections, comparisons, and restoring divisions gives shared pointwise depth $O(p^2)$ and width $O(p)$. The twelve scalar extractions, when present, add only $O(h+1)$ depth and fit that width. Attention's exact integer evaluator is charged as work below and is not hidden in this small hybrid depth.

Individual coefficients use $O(p)$ bits. Boolean coefficients are bounded constants, scalar-extraction coefficients use $O(h+1)$ bits, and output dyadics use $O(b+1)$ bits. The shared pointwise circuit has $O(p^2)$ active gates. Explicitly holding $O(p)$ live rails through all layers gives $O(p^3)$ sparse gate/hold records. Their addresses and coefficients cost at most $O(p)$ bits each, hence $O(p^4)$ stored sparse parameter bits. The dense alternative has $O(p^2)$ layers, $O(p)$ width, and therefore $O(p^4)$ coefficient slots including zeros and biases. Storing all coefficients costs $O(p^5)$ bits. Constantly many head-projection matrices fit these bounds. Finite evaluator code and bounded-word templates cost at most $O(p^4)$ bits and are included, not left as an external oracle.

Masks, public positions, and input. Binary input acquisition costs $M+2$ bits. The synthetic zero is public. For positions, query type, half flags, row boundaries, and masks, the two complete choices are:

1. *Active-key lists.* The first two heads each contain n stored keys. The third contains $n-1+M$ keys. Thus the exact number of stored key occurrences is $3n-1+M$. Each key and row boundary uses $O(\ell_a)$ bits, giving $O(n\ell_a)$ mask/list storage, including positions and flags. Enumerating indices, comparisons, routes, and row boundaries costs at most $O(n\ell_a^2)$ bitwork. Each stored route is read and charged during execution. This is explicitly paid sparse structural masking, not free addressing.

2. *Literal dense masks.* Three n-by-n masks cost exactly $3n^2$ mask bits, in addition to $O(n\ell_a)$ metadata. Generating and writing them costs at most $O(n^2\ell_a)$ bitwork. A dense online scan costs at most $O(n^2 p)$ even though many permitted-key sets are singletons. Converting active lists to this representation pays these dense generation and storage costs.

Finite attention arithmetic. There are $O(n)$ allowed key occurrences in the active-list version. Each copied value is a bit. For the uniform head, each of twelve rails per key uses an $O(p)$-bit addition, and every token performs twelve $O(p^2)$ rounding divisions. Thus the fixed-mask finite attention evaluation costs $O(np^2)$ elementary bitwork, including copies, projections, and clearing. No logarithm, exponential, or source-specific cache is generated. Dense-mask scanning adds the cost just stated. Controllers and word addresses occupy $O(p)$ bits; the larger complete runtime bounds below also allow a full $O(p)$ overhead per addressed record lookup.

Two complete online implementations. Sparse execution of all pointwise maps, including every live hold, costs $O(np^3)$ elementary bitwork. Combining this with active-list attention, then charging up to $O(p)$ per sparse record access, gives the conservative unconditional sparse/sparse online bound

$$T_{\text{sparse}} = O(np^4). \tag{F.21}$$

Its stored parameters and code are $O(p^4)$ bits, its masks and public metadata are $O(n\ell_a)$ bits, and its observed input costs $M + 2$ bits. Dense coefficient counts are not substituted into this runtime.

For the dense-mask/dense-circuit alternative, coefficient-bit operations are charged even for zero terms. Dense QKV projections, all-token shared maps, mask scans, and word operations give the conservative bound

$$T_{\text{dense}} = O(n^2p^2 + np^6). \tag{F.22}$$

Its parameters and code cost $O(p^5)$ bits, with masks and metadata $O(n^2 + n\ell_a)$ bits. The quadratic term is a safe allowance for this dense implementation, not a lower bound or an assertion that the sparse implementation performs quadratic work.

Activations, cleanup, and total online peak. With bit rails and packed words, the working activation arrays require $O(np)$ bits. A conservative scalar-channel implementation allocates $O(p)$ physical bits to each of $O(p)$ live channels per token, hence $O(np^2)$ bits. These allowances include Q/K/V arrays, feature residuals, lag holds, twelve pre-clearing bank words per token, endpoint holds, controller, and arithmetic scratch. Input costs $O(n)$ additional bits; output costs $O(b + 1)$ packed bits, or $O((b + 1)p)$ in the conservative scalar allocation. The explicit $+1$ covers $b = 0$ and the numerator 2^b. Initializing, overwriting, and releasing these arrays is charged in the online bounds. Only the latest live layer is retained: a training tape or all-layer retention would be an additional cost.

Total online peak is the sum of simultaneously resident program and parameters, masks and metadata, observed input, QKV and residual arrays, banks, controller and scratch, and output. With conservative scalar allocation it is bounded by

$$S_{\text{sparse}} = O(p^4 + n\ell_a + np^2), \tag{F.23}$$

$$S_{\text{dense}} = O(p^5 + n^2 + n\ell_a + np^2). \tag{F.24}$$

The $12(h + 1)$ bank bits and two endpoint bits are only the cut payload, not these total-memory quantities. At h of order $\frac{1}{2}\log_2 M$, exact accumulators, addresses, and public horizon words still require order $\log M$ bits, and the resident input and paid mask storage remain.

Offline construction and peak storage. Generate the ripple circuits, fixed division schedules, muxes, scalar extractions, and all identity holds in $O(p^5)$ conservative bitwork with $O(p^4)$ sparse scratch/record storage, and write the $O(p^4)$ sparse program bits. Sorting, if used, involves $O(p^3)$ records of $O(p)$ bits and $O(\log p)$ comparison overhead, which fits the same work allowance. The

dense representation additionally writes $O(p^5)$ coefficient bits, including zero slots, within $O(p^5)$ bitwork. Add $O(n\ell_a^2)$ work for sparse lists or $O(n^2\ell_a)$ work for dense masks and their public metadata. Accordingly, safe full construction work bounds are

$$O(p^5 + n\ell_a^2) \quad \text{(sparse)}, \qquad O(p^5 + n^2\ell_a) \quad \text{(dense)}. \tag{F.25}$$

Offline peak includes partially generated stored program and masks plus live generation scratch. It is bounded by the corresponding total in (F.23) or (F.24); retaining both construction and execution workspaces adds their sum, of the same order. There is no hidden training phase, source-specific compilation, or external arithmetic oracle.

A separate all-Boolean/ReLU unrolling. To remove the finite-attention primitive, select one complete representation and unroll its fixed public bit-operation schedule. Let T be that representation's conservative online bitwork and S its total live bit storage, including input, program/constants, masks, and scratch. Every conditional data operation is already a Boolean mux; loop lengths and permitted routes are public. Replace each elementary bit operation by (F.14)–(F.15), and at every instruction explicitly hold every other live rail. Induction on instructions proves that the resulting network state equals the original finite evaluator's state. The final dyadic decode is also retained and charged.

This produces a *PURE* Boolean/ReLU network, with no finite-attention arithmetic primitive, having the valid bounds

$$\begin{aligned} \text{depth} &= O(T+p), & \text{width} &= O(S+p), \\ \text{sparse gates and holds} &= O((T+p)(S+p)), \\ \text{dense parameter bits} &= O(p(T+p)(S+p)^2), \\ \text{dense bitwork} &= O(p^2(T+p)(S+p)^2). \end{aligned} \tag{F.26}$$

For example, one may use $T = O(np^4)$ and $S = O(p^4 + n\ell_a + np^2)$ from the sparse hybrid. The sparse-record description can conservatively be written with $O(p(T+p)(S+p))$ bits: all relevant indices have $O(p)$ bits, since $\log n \le p$ and $\log p \le p$. Writing and evaluating the PURE representation incurs its own displayed storage and work costs. These large unrolling bounds do not inherit the hybrid depth $O(p^2)$ or per-token width $O(p)$. The small hybrid shape therefore must never be presented as the shape of the fully expanded finite evaluator.

F.8 Observation comparisons and the classical baseline

The count-only obstruction has an information scope. Theorem 3.1, specialized to $r = 2$, gives an optimal budget-blind competitive factor between

$$2^{-30}\left(1 + \log\frac{M+2}{2}\right) \quad \text{and} \quad 2^{20}\left(1 + \log\frac{M+2}{2}\right)$$

for exact triple counts plus both endpoint pairs, at supplied $1/M$. The lower tests the same map across budgets; it is not a logarithmic penalty at every individual budget. Counts plus only the protected pair, and any rounded versions of those observations, are weaker and inherit the obstruction. The present half-position statistics change the observation, so that comparison supplies neither an architecture-wide lower bound nor a neural depth separation.

A one-pass counter on equal positional information. On the same attacked word, a classical program keeps four exact triples of counters (J_u, C_u, D_u) and a two-bit sliding window. When symbol Z_{i+2} arrives, the window identifies $u = (Z_i, Z_{i+1})$: increment J_u, increment C_u if $Z_{i+2} = 1 - Z_i$, and increment D_u if $i < \lfloor M/2 \rfloor$. Then shift the two-bit window. After the last symbol it contains the protected endpoint. Thus the program computes exactly (F.2) in one pass, without knowing k.

Its counters, loop index, and sliding window use $O(\log M)$ auxiliary bits, with $O(M \log M)$ conservative bitwork for the scan. On the exact counters, the same positive-denominator check, gate, and finite output division give the risk in (F.5) with $\eta = 0$. If output precision is charged, the division uses $O((\log M + b + 1)^2)$ active bitwork and $O(\log M + b + 1)$ live bits, with additional holds if fully layered. Alternatively, round the exact counters once at the end using (F.12). This reproduces the attention bank coordinate for coordinate, so it is also an equal-bank baseline, not merely an asymptotically comparable estimator.

The input models must be matched. If the word is resident, charge the same $M + 2$ input bits to the counter and the attention program. If a read-once input stream is allowed, state that permission explicitly; do not compare the counter's auxiliary workspace with the attention program's batch total memory. The counter has no attention mask-storage cost. Consequently no neural statistical or computational advantage over equal positional information is asserted. Sparse masks make the proposed realization cheaper than generic dense attention evaluation, but do not surpass this sequential baseline.

What changes relative to the earlier finite bank. The construction of BCJ001 [1] already supplies eight full-triple and four half-row normalized coordinates, public half-position flags, uniform attention, endpoint acquisition, an isolated finite observation, and a paid finite sequential evaluator with optional full unrolling. It also already distinguishes fractional bank precision from arithmetic word width and allows a final rounded probability. Its approximate-gate, quantization, noise, and source-edit results establish the central finite-bank precision phenomenon in that earlier interface.

The present implementation makes a different, explicit choice at several points. Two singleton heads acquire the lagged bits and endpoint. The local Boolean circuit emits four direct $J/C/D$ triples. A zero-score head uses exact accumulation followed by one nearest rounding, rather than the earlier evaluator's nested floors. The numeric fault is a canonical replacement *after* that rounding, rather than an additive analog change before a final floor. Bit circuits, scalar-grid extraction, restoring division, and dyadic output give the specified two-layer, three-head hybrid with growing pointwise depth. Before rounding, the full-triple and direct-row representations are linearly interconvertible; after separately applied rounding they need not define the same statistical experiment. Neither port theorem, especially a port-noise lower bound, transfers merely because both banks have twelve labels.

Uniform indicator aggregation, positional copying, and finite arithmetic compilation are standard ingredients [7, 17, 19, 20]. The construction here uses exactly accumulated indicators followed by rounding of their aggregate, not rounding each attention weight $1/M$ onto the h-grid. Results about rounding attention probability vectors therefore concern a different numerical operation. The explicit singleton structural masks likewise must not be confused with dense-causal-only attention or finite-temperature approximations to copying. No conclusion is asserted about training, learnability, minimal head count, total-precision optimality, architecture-wide lower bounds, pure-network depth advantages, empirical performance, or priority for these standard mechanisms.

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